



SACC

2020 中国系统架构师大会

SYSTEM ARCHITECT CONFERENCE CHINA 2020

架构融合 云化共建

LIVE 2020年10月22日 - 24日网络直播

架构融合
云化共建

央行数字货币架构研究

朱玮



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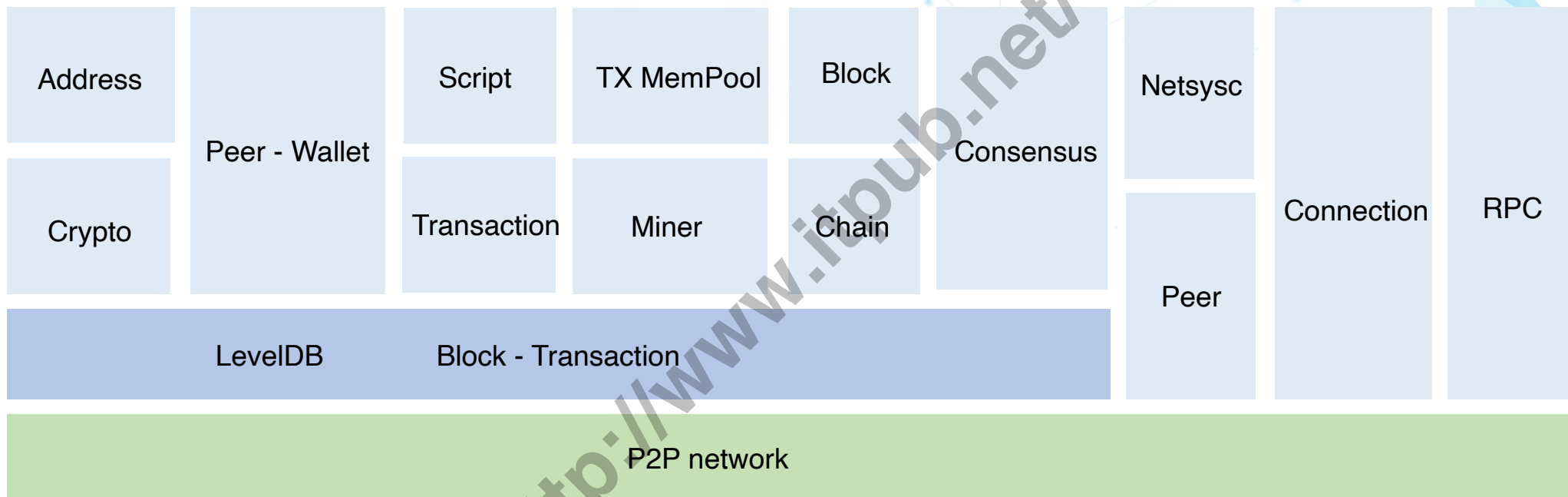
1

区块链架构

Bitcoin 架构

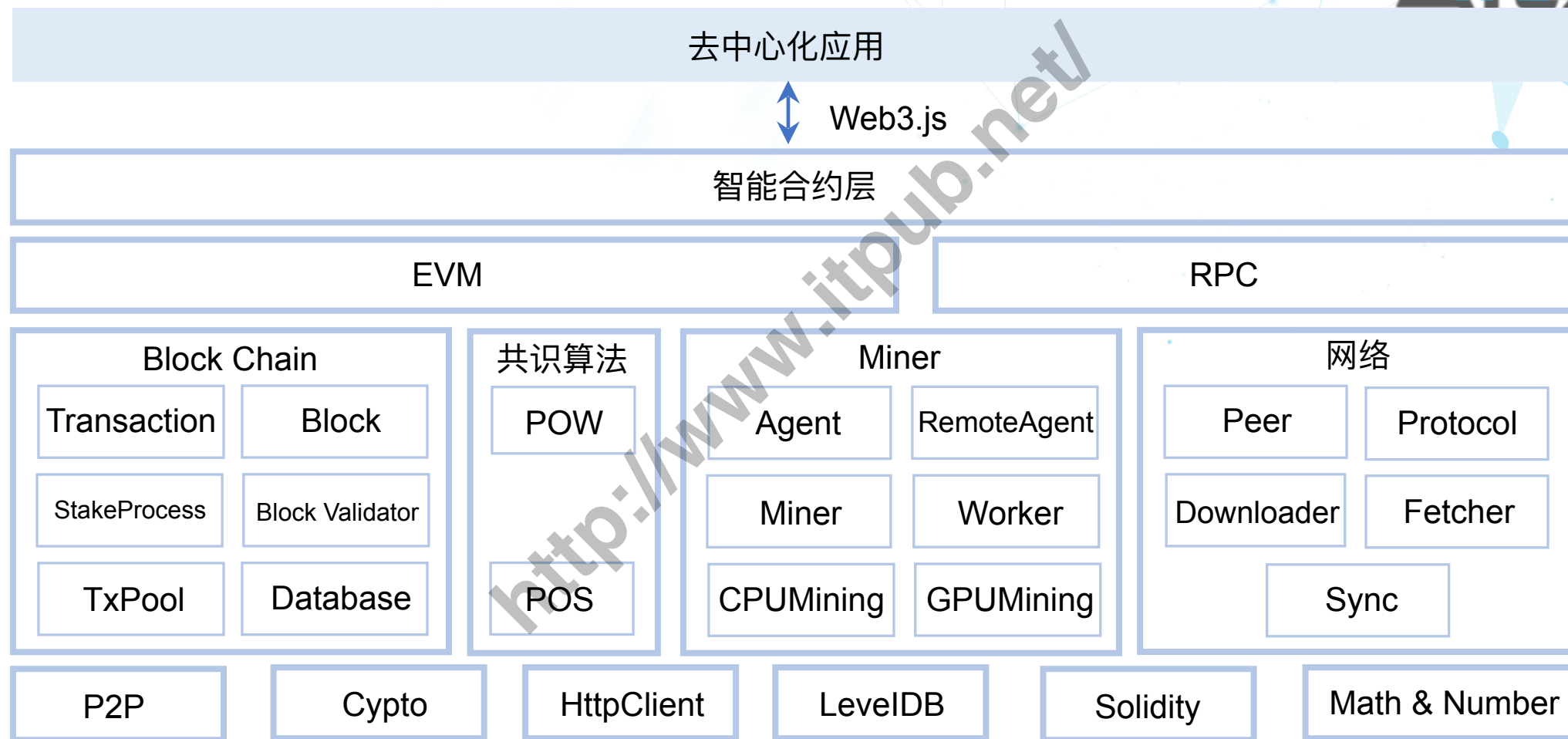
架构融合

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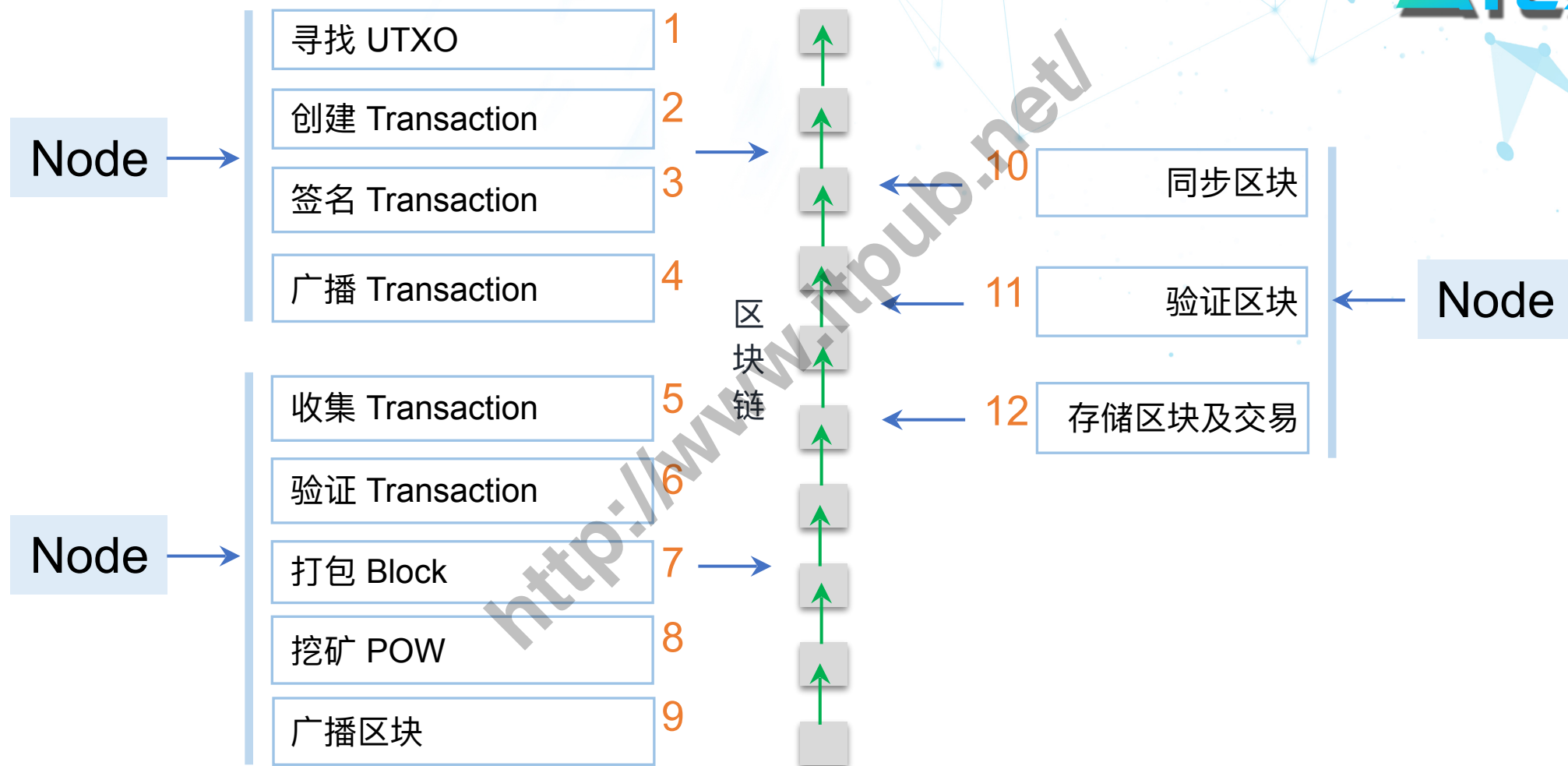


一种 P2P 的电子现金系统

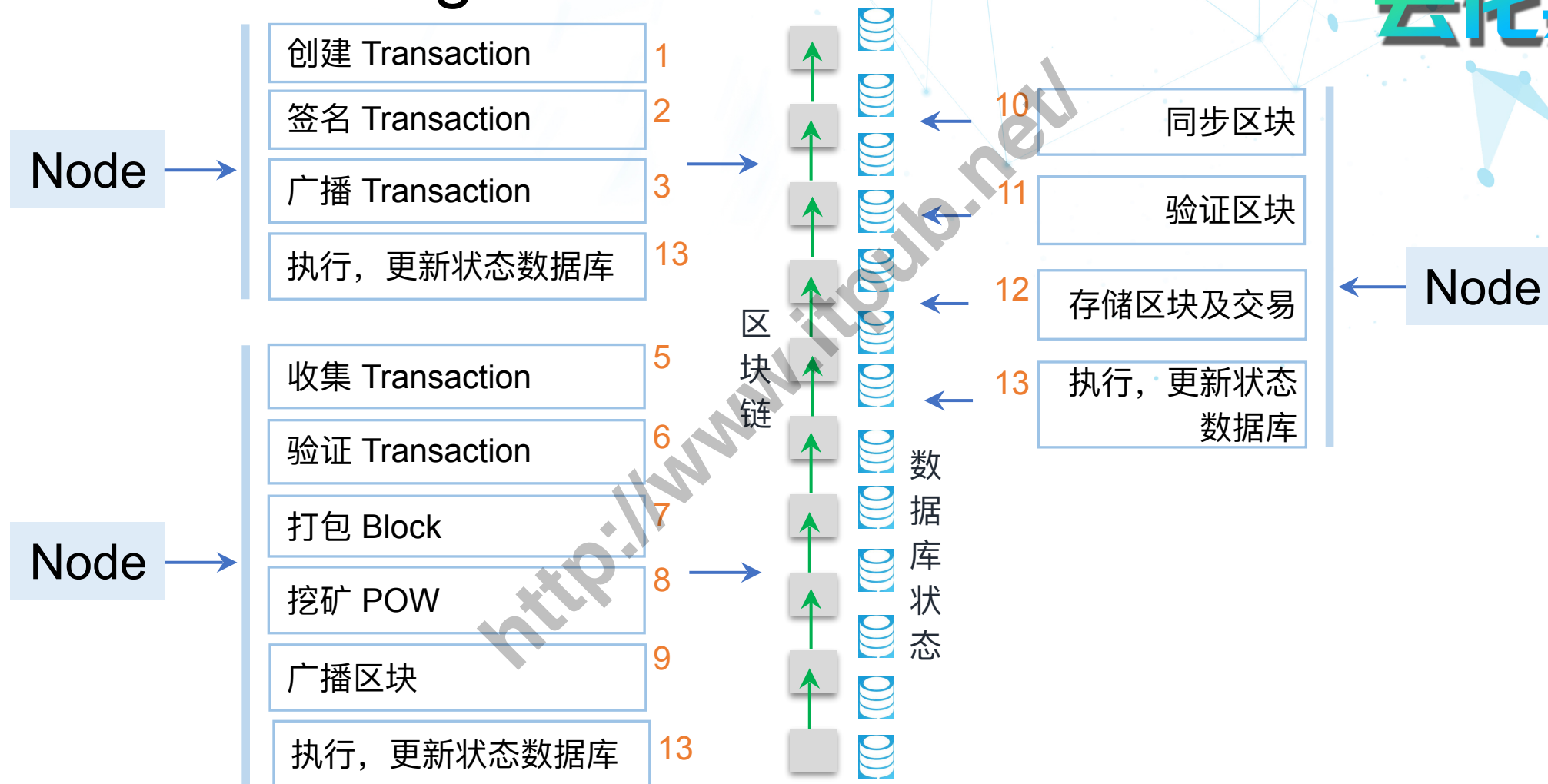
以太坊架构



Bitcoin Logic



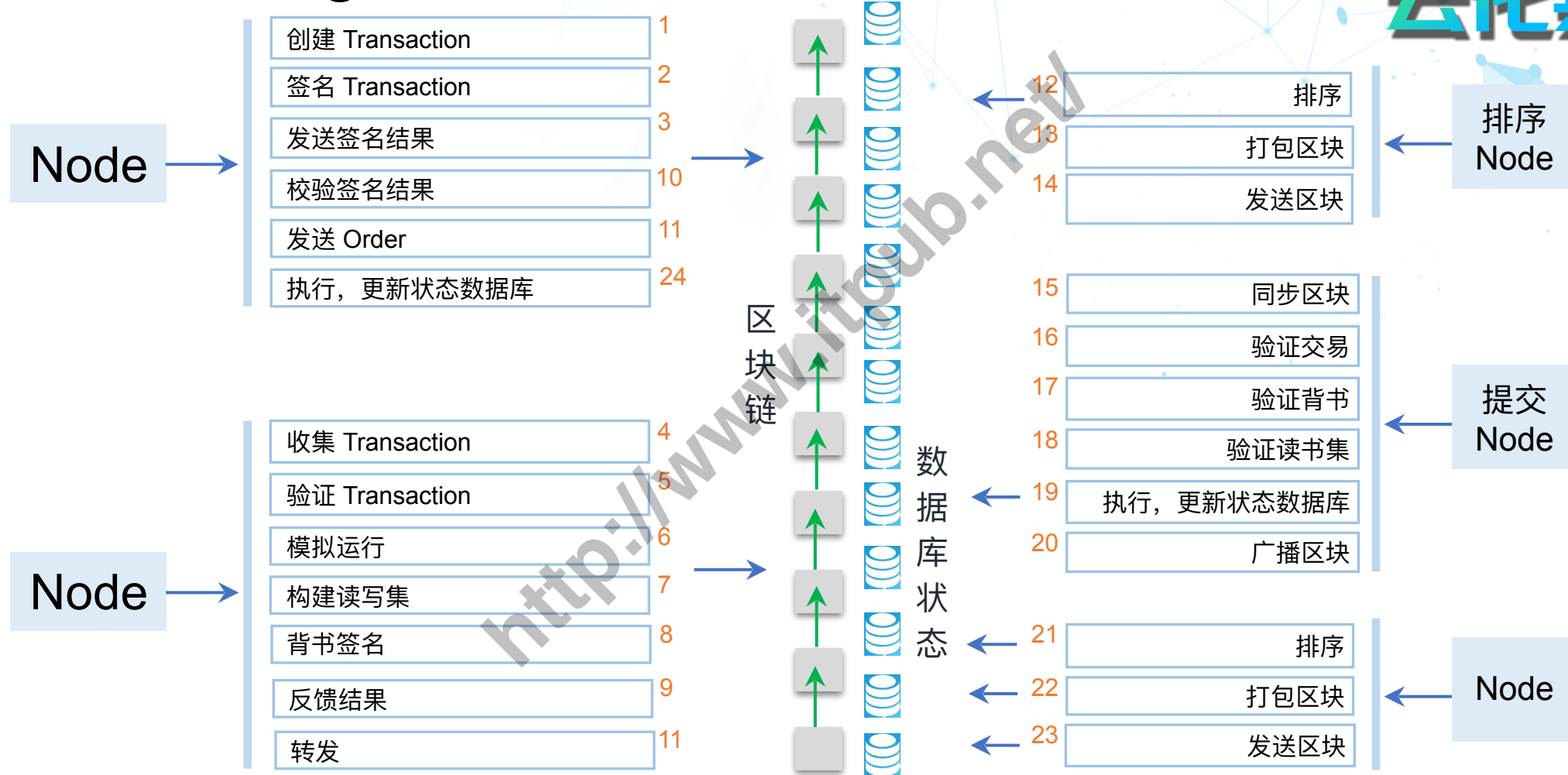
Ethereum Logic



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Fabric Logic



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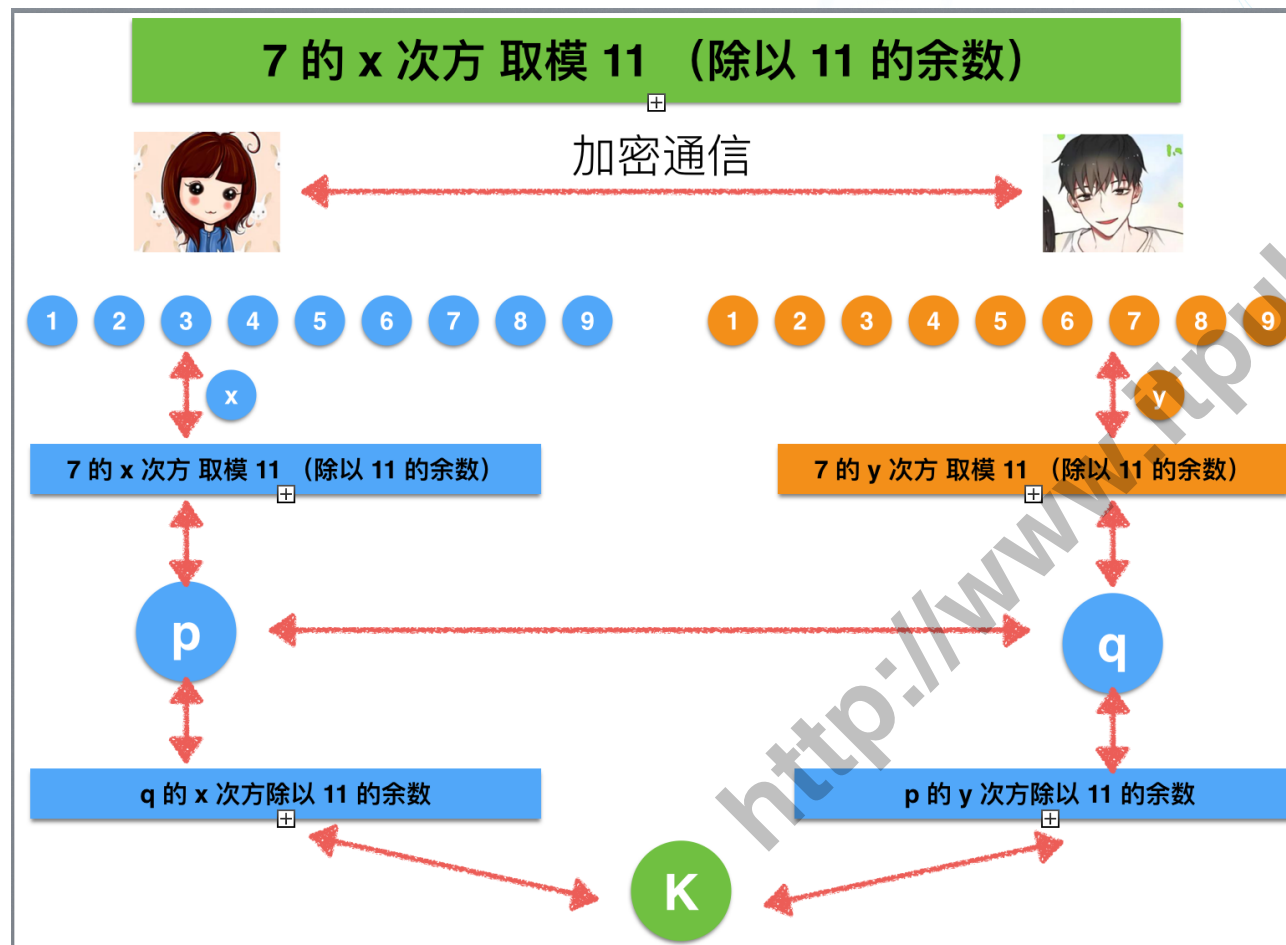
2

数字货币的历史

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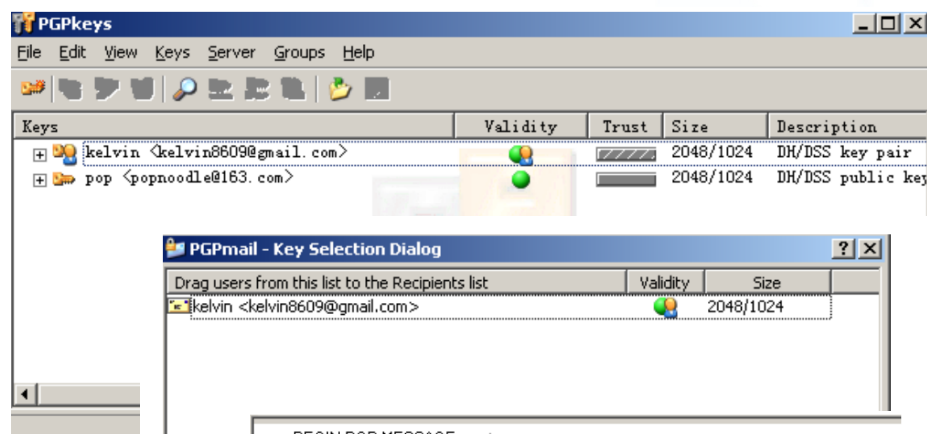
DH 算法



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PGP 邮件加密

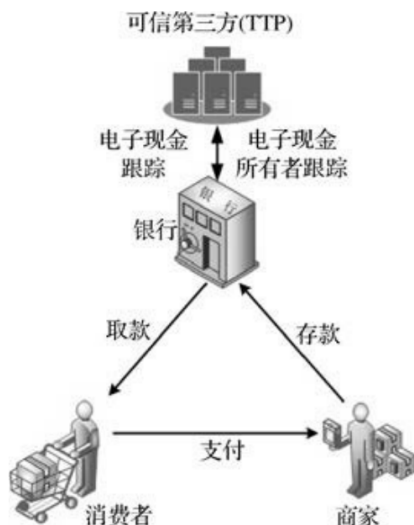


```
-----BEGIN PGP MESSAGE-----  
Version: PGP 8.0  
  
qANQR1DBwU4DJM+IWRHioFkQCACfgmOrydpqCA5NK39DTqX8+1LZsdU5OgBg8N6  
O7bvEFRIf1ZKFWF9NBPd6QrwjUUVYumQMmhYAXqNCSwZyD4sQU8U+7x8+7BGV  
RXoToedrd63gbsGmsfKUBgjQku7kyKCAKA3w1ld1kNIEumdiariq/4RLwodK11hP  
YEXscE61mPJQoRe+ZRQEPbsqp2nt2TRY4y2SuV75BE3ZdTcwMqQYb5YP3SGnmqb  
JHhrZjuUgjk5jDSANjUgBHtV0wMfsSUxRXsRwyA5YnqvNSmHYaZC4JAK7Hhbv  
CndvEh5SSqCOSU0+mzj  
2SB06bRE09apH1deUN.  
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+AXa9NAIdQXlZJt69cmKE  
Rk7IO2t3Xwa5XcUWdn6k  
yRT++4xtnJDKMkoR5cYjY  
gQrk5MF5yb+d0Yuj9TfOIS  
BEXz4E8JeQamsrWCz5z  
ga3QhaMas8E/qzDZO/kR  
saJk0hQ76k4QSMoLfmf  
Emlu4NU4bw==  
=dRjk  
-----END PGP MESSAGE-----
```

Text Viewer

```
*** PGP SIGNATURE VERIFICATION ***  
*** Status: Good Signature ***  
*** Signer: kelvin <kelvin8609@gmail.com> (0x7410B0C5) ***  
*** Signed: 2009-12-24 13:22:20 ***  
*** Verified: 2009-12-24 15:21:35 ***  
*** BEGIN PGP DECRYPTED/VERIFIED MESSAGE ***  
我们写一封测试信，内容如下，当写完信的内容以后，将信的内容全选，然后选择剪切。  
如图  
*** END PGP DECRYPTED/VERIFIED MESSAGE ***
```

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eCash 电子现金

简单的展开一下，eCash 是用数字货币实现了纸钞的防追踪性。王美丽到银行取钱，银行从她的账户上扣除 1 万元，同时给她 1 万元纸钞。王美丽持 1 万元纸钞在商场购物，店家并不查验王美丽的身份，也并不记录王美丽的购物信息，店家只关心纸钞的真实性。之后，店家把 1 万元纸钞存进银行，银行也并不知道这 1 万元来自于

电子现金支付旨在网络上重建基于现金型交易功能的作用，它可以使用密码技术和盲签名技术实现完全匿名，从而保护用户的消费信息。自从 1982 年 Chaum 最早提出一个在线的基于 RSA 盲签名完全匿名电子现金方案以来^[7]，研究者们提出了各种各样的方案。

1988 年，Chaum、Fait、Naor 利用切割、选择和 RSA 盲签名技术提出了一个在线的匿名电子现金方案^[8]；1991 年，Okamoto、Ohta 采用二叉树结构表示电子现金和利用切割选择技术的可分电子现金方案^[9]；1992 年，由 Brands 最早利用限制性盲签名提出了一个离线

的完全匿名电子现金方案，这是迄今为止效率最高的方案之一，1993 年又进行了改进^[10]；1995 年，Stadler、Brickel 分别提出公平电子现金方案，除具备一般电子现金的功能外，还可跟踪电子现金和所有者跟踪协议^[11]。同年 Brickel、Ghemmel、Kravitz 引入基于可信第三方(Trusted Third Party, TTP)的跟踪^[12]。1996 年，Frankel、Tsionis、Yung 基于间接证明技术提出了一个公平电子现金方案^[13]；1997 年，Davida、Frankel 和 Tsionis 基于零知识证明技术对上述方案进行了改进^[14]，1998 年，Frankel、Tsionis 和 Yung 利用同构加密的思想分两次对上面方案进行了改进^[15]。1998 年，Anna、Zulfikar 首先提出了基于群盲签构造一个具有多个银行参与发行的完全匿名电子现金方案^[16]；1999 年，杨波等提出了选用灵通卡的公平电子现金方案^[17]。

用户 Alice 在银行 B 开户，Alice 在 PC 上安装一个 eCash 客户端。当 Alice 需要取钱的时候：

1. Alice 发送一个请求 x 给银行 B，该请求是密文 $c(x)$ ，银行 B 无法解密
2. 银行 B 获得 Alice 请求后，扣减 Alice 账户，并对请求的密文消息执行“盲签名”- $s(c(x))$
3. Alice 获得签名后的 $s(c(x))$ 后，进行脱盲运算得到 $c'(s(c(x))) = s(x)$ ，也就是对请求 x 的银行签名
4. Alice 支付 $s(x)$ 给 Bob
5. Bob 将 $s(x)$ 通知银行，银行将一个美元存入 Bob 的账户

架构融合 云化共建

b-money

Wei Dai

November 1, 1998

I am fascinated by Tim May's crypto-anarchy. Unlike the communities traditionally associated with the word "anarchy", in a crypto-anarchy the government is not temporarily destroyed but permanently forbidden and permanently unnecessary. It's a community where the threat of violence is impotent because violence is impossible, and violence is impossible because its participants cannot be linked to their true names or physical locations.

Until now it's not clear, even theoretically, how such a community could operate. A community is defined by the cooperation of its participants, and efficient cooperation requires a medium of exchange (money) and a way to enforce contracts. Traditionally these services have been provided by the government or government sponsored institutions and only to legal entities. In this article I describe a protocol by which these services can be provided to and by untraceable entities.

I will actually describe two protocols. The first one is impractical, because it makes heavy use of a synchronous and unjammable anonymous broadcast channel. However it will motivate the second, more practical protocol. In both cases I will assume the existence of an untraceable network, where senders and receivers are identified only by digital pseudonyms (i.e. public keys) and every messages is signed by its sender and encrypted to its receiver.

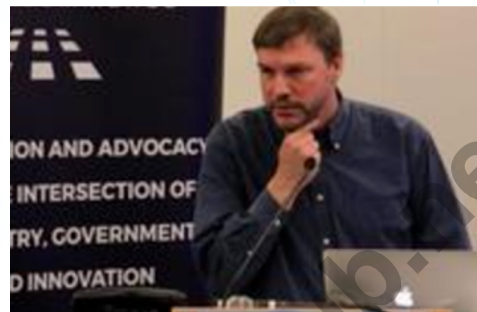
Bit gold

A long time ago I hit upon the idea of bit gold. The problem, in a nutshell, is that our money currently depends on [trust in a third party](#) for its value. As many inflationary and hyperinflationary episodes during the 20th century demonstrated, this is not an ideal state of affairs. Similarly, [private bank note issue](#), while it had various advantages as well as disadvantages, similarly depended on a trusted third party.

[Precious metals and collectibles](#) have an unforgeable scarcity due to the costliness of their creation. This once provided money the value of which was largely independent of any trusted third party. Precious metals have problems, however. It's too costly to assay metals repeatedly for common transactions. Thus a trusted third party (usually associated with a tax collector who accepted the coins as payment) was invoked to stamp a standard amount of the metal into a coin. Transporting large values of metal can be a rather insecure affair, as the British found when transporting gold across a U-boat infested Atlantic to Canada during World War I to support their gold standard. What's worse, you can't pay online with metal.

Thus, it would be very nice if there were a protocol whereby unforgeably costly bits could be created online with minimal dependence on trusted third parties, and then securely stored, transferred, and assayed with similar minimal trust. Bit gold.

My proposal for bit gold is based on computing a string of bits from a string of challenge bits, using functions called variously "client puzzle function," "proof of work function," or "[secure benchmark function](#)". The resulting string of bits is the proof of work. Where a [one-way function](#) is prohibitively difficult to compute backwards, a secure benchmark function ideally comes with a specific cost, measured in compute cycles, to compute backwards.



B-Money & Bit Gold

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防范垃圾邮件

Hashcash

Hashcash is a proof-of-work algorithm, which has been used as a denial-of-service counter measure technique in a number of systems.

A hashcash stamp constitutes a proof-of-work which takes a parameterizable amount of work to compute for the sender. The recipient (and indeed anyone as it is publicly auditable) can verify received hashcash stamps efficiently. Hashcash was invented by [Adam Back](#) in 1997.

At this point it is most widely used as the [bitcoin](#) mining function.

The email anti-spam tool, like the proof-of-work algorithm, is also called hashcash and is used to create stamps to attach to mail to add a micro-cost to sending mail to deter spamming. The main use of the hashcash stamp is as a white-listing hint to help hashcash users avoid losing email due to content based and blacklist based anti-spam systems.

Hashcash source code includes a library form, and also the algorithm is extremely simple to code from scratch with the availability of a hash library. Verification can be done by a human eye (count leading 0s) even with availability of common preinstalled command line tools such as sha1sum. The algorithm works with a cryptographic hash, such as SHA1, SHA256 or coming SHA3 that exhibits 2nd-preimage resistance. Note that 2nd-preimage resistance is a stronger hash property than the collision resistance property.

- Hashcash [FAQ](#).
- Hashcash [format info](#)
- Hashcash Paper Aug 02 - "[Hashcash - A Denial of Service Counter-Measure](#)" (5 years on), Tech Report, *Adam Back*
- The site is slowly moving over to this new domain, some of the content is still here: <http://www.cypherspace.org/hashcash/>

哈希现金



Reusable Proofs of Work

by Hal Finney
(hal dot finney at gmail dot com)

[News](#) [What is This?](#) [Theory](#) [Security](#) [Try it Out!](#) [FAQs](#) [Presentation](#) [Download](#)

(The RPOW project is now terminated. These pages are maintained for historical purposes.)

The RPOW system provides for proof of work (POW) tokens to be reused. A POW token is something that takes a relatively long time to compute but which can be checked quickly. RPOW uses [hashcash](#), which are values whose SHA-1 hashes have many high bits of zeros.

Normally POW tokens can't be reused because that would allow them to be double-spent. But RPOW allows for a limited form of reuse: sequential reuse. This lets a POW token be used once, then exchanged for a new one, which can again be used once, then once more exchanged, etc. This approach makes POW tokens more practical for many purposes and allows the effective cost of a POW token to be raised while still allowing systems to use them effectively.

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层次三： 数字化价值记录 - 比特币

是否在发行环节摆脱了
银行？非债！

层次二： 现实价值关系的数字化记录 - 稳定币&央行
数字货币

是否支付和清算环节
摆脱了银行？

层次1.5： 现实价值的数字化记录 - eCash

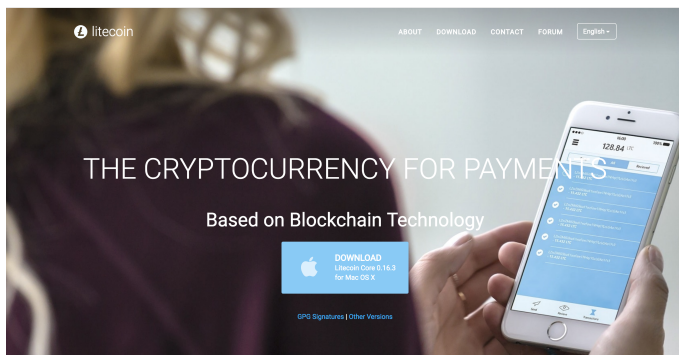
是否在支付环节摆脱了
银行？

层次一： 现实价值的数字记录 - 网银支付宝

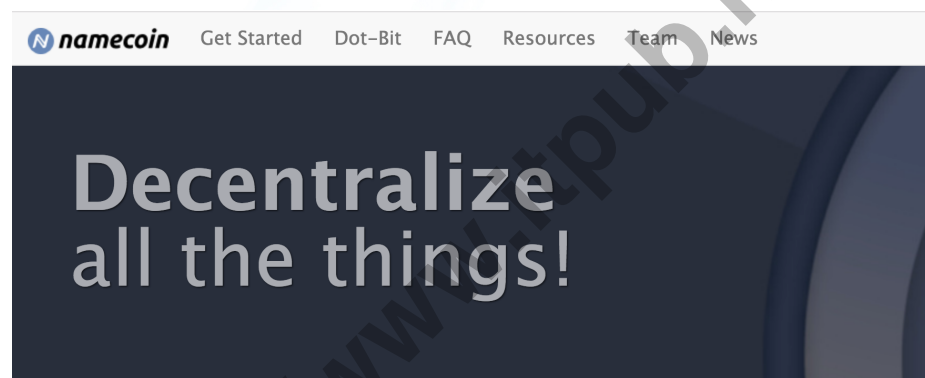
是否摆脱了物理实体？

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莱特币



域名币



架构融合 云化共建

1EXoDusjGwvnjZUyKkxZ4UHEf77z6A5S4P



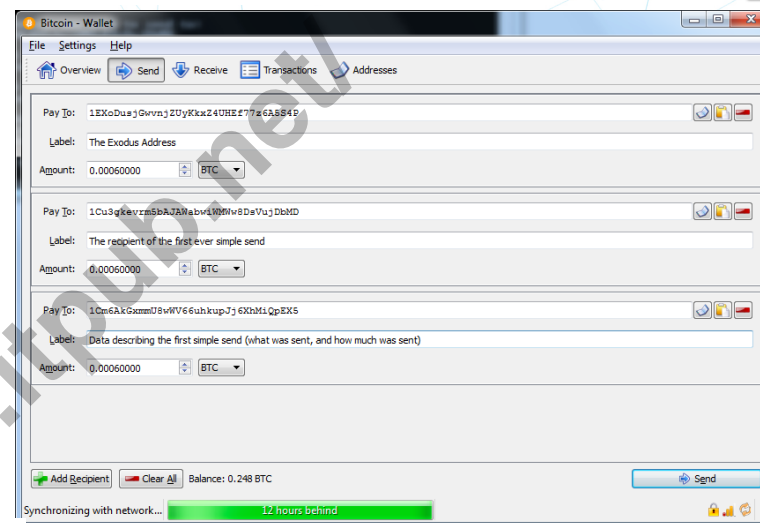
用户定义货币：任何人能在Mastercoin网络上创造他们自己的货币。

去中心化交易所：Mastercoin网络自己充当一个支持在任何两种在Mastercoin网络里的货币之间进行兑换的全功能交易所。任何人能够在区块链上下订单来兑换一些货币，其他任何人能够接受这个订单并且交易是全自动的，没有订单告示的进一步介入。

区块链上的价格提供：可信的组织能发布能被用在Mastercoin脚本上的价格数据（例如，一盎司黄金兑美元的价格）。

区块链上的赌博：与另一方登记一个有关已知价格在未来某一特定时刻超出或低于某一价格的赌博是完全有可能的。它本质上允许针对货币对的杠杆投机和对冲，这样用户能够在不顾来自比特币或Mastercon价格波动性的高度货币风险的情况下向Mastercoin网络存入资金。

储蓄地址：来自一个储蓄地址的交易能在N天内被一个“保护人地址”撤销（N对每个地址都是一个定值）。它本质上为大资金储蓄账户提供一个额外的安全保证。



万事达币

架构融合 云化共建

下一代智能合约和去中心化应用平台

```
pragma solidity ^0.4.21;

contract Coin {
    address public minter;
    mapping (address => uint) public balances;

    // Events allow light clients to react on
    // changes efficiently.
    event Sent(address from, address to, uint amount);

    // This is the constructor whose code is
    // run only when the contract is created.
    function Coin() public {
        minter = msg.sender;
    }

    function mint(address receiver, uint amount) public {
        if (msg.sender != minter) return;
        balances[receiver] += amount;
    }

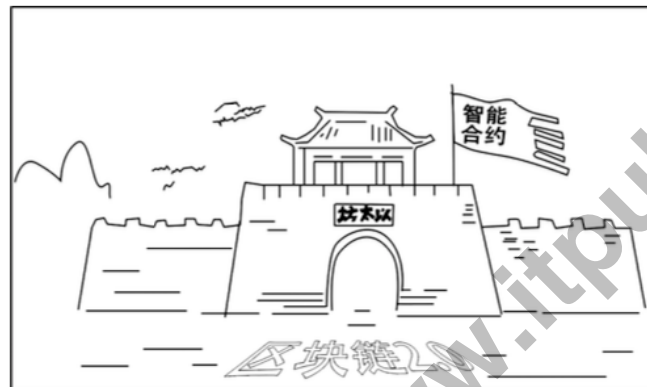
    function send(address receiver, uint amount) public {
        if (balances[msg.sender] < amount) return;
        balances[msg.sender] -= amount;
        balances[receiver] += amount;
        emit Sent(msg.sender, receiver, amount);
    }
}
```

创建一个叫 "Coin" 的合约
定义公共变量 Minter, 用来存放以太坊地址
定义公共变量 balances, mapping 类型

初始化 Coin 这个合约
将初始化的以太坊地址, 设置为 minter

定义功能 mint, 用来发行代币
代币发行只能由 minter 地址执行, 其他地址执行则出错
balances[receiver] += amount; 发行代币到 receiver 地址

定义功能 send, 用来转移代币
如果要转移代币的地址余额不足, 则报错
balances[msg.sender] -= amount; 转移代币的地址减去相应数量的余额
balances[receiver] += amount; 接受代币的地址加上相应数量的余额
记录日志



据 etherscan.io 网站在 2018 年 6 月 12 日的数据显示, 以太坊上的 ERC 20 代币智能合约共计 90738 种。另据可靠数据统计, 2018 年第一季度, 在以太坊上筹集的资金就高达 65 亿美元。

2018 年 6 月 12 日, 以太坊上共约 3700 万账户地址, 每日交易量达 150 万笔。以太坊全球共有 17000 个节点, 开发者社区共有 25 万名开发者。

据 stateofthedapps.com 网站在 2018 年 6 月 12 日的数据显示, 以太坊上共部署了 1600 个应用。

据 dappradar.com 网站 2018 年 6 月份统计, 以太坊上的 Dapp 应用, 以游戏种类占比最大, 达到 70%; 赌博类次之, 为 10%; 交易市场为 2.7%; 其它类型共约 17 %。

Ethereum

架构融合 云化共建

EOS 并不信奉“代码即法律”这句话，而是按照代议制体制，建设了一套结合了线下和线上的治理模式。所以，EOS 社区有宪法，是社区判断善恶的根本标准。

在 EOS 上的智能合约，要用李嘉图合约（Ricardian Contract）格式来用文字描述合约意图，并且指定发生纠纷后的仲裁方式。

在 EOS 的哲学中，并非“代码即法律”，而是“代码背后的人类意志才是法律”

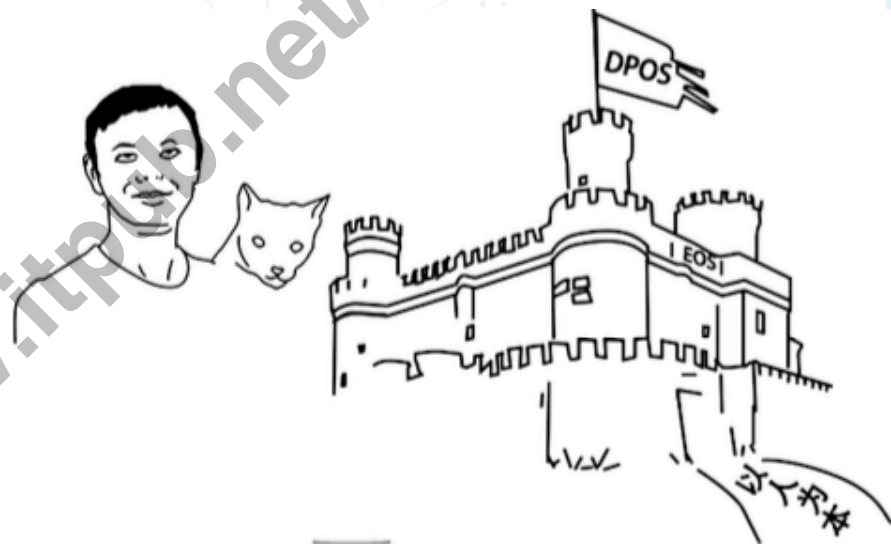


图 6-4 以人为本的 EOS

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3

各国央行数字货币

Fedcoin

J.P. Koning



Hi-Res Image

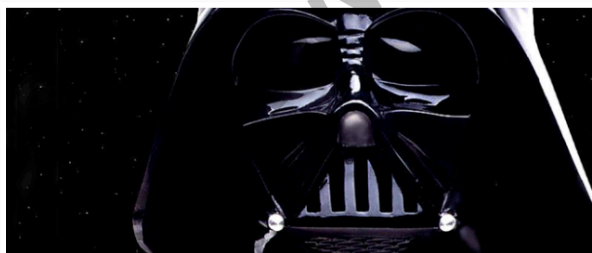
Contributor, Sound Money Project

J.P. Koning is a financial writer and blogger with interests in monetary economics, economic history, finance, and fintech. He has worked as an equity researcher at a Canadian brokerage firm and a financial writer and publisher at a large Canadian bank. More recently, he has written several papers for R3, a distributed ledger company, on the topics of central bank cryptocurrency and cross border payments. He founded the popular blog [Moneyness](#) in 2012. He designs economics and financial wallcharts at [Financial Graph & Art](#).

Koning earned his B.A. in Economics from McGill University.

Sunday, October 19, 2014

Fedcoin



Recent posts by [Adrian Hope Baille](#) and [Sina Motamedi](#) have got me thinking again about the idea of the Federal Reserve (or any other central bank for that matter) adopting bitcoin technology. [Here's](#) an older post of mine on the idea, although this post will take a different tack.

The bitcoin ethos enshrines the idea of a world free from the totalitarian control of central banks. So in exploring the idea of Fed-run bitcoin-style ledger, I realize that I run the risk of being cast as Darth Vader (or even "yikes" the Emperor) by bitcoin true believers. [So be it](#). While I do empathize with the bitcoin ideal—I support freedom in banking—I rank the importance of *bitcoin-as-product* above *bitcoin-as-philosophy*. And at the moment, bitcoin is *not* a great product. While bitcoin has many useful features, these are all overshadowed by the fact that its price is too damn volatile for it to be taken seriously as an exchange medium. This volatility arises because bitcoin lacks a fundamental value, or anchor, a point that I've written about [many times in the past](#). However, there is one way to fix the crypto volatility problem...

The big losers: banks

Fedcoin has the potential to tear down the private banking system. Interest yielding Fedcoin would be able to do everything a bank deposit could do and more, and all this at a fraction of the cost. As the public shifted out of private bank deposits and into Fedcoin, banks would have to sell off their loan portfolios, the entire banking industry shrinking into irrelevance.

One way to prevent this from happening would be for the Fed to make an explicit announcement that any bank could be free to create its own competing copy of Fedcoin, say *WellsFargoCoin*. Like the Fed, Wells Fargo would promise to offer two-way convertibility between its deposits/cash/Fedcoin and WellsFargoCoin at a rate of 1:1 to ensure that the price of its new ledger entries were well-anchored. The bank could then implement features to compete with Fedcoin such as higher interest rates or complimentary financial services. Even as Wells Fargo's deposit base steadily shrunk due to technological obsolescence, its base of WellsFargoCoin liabilities would rise in a compensatory manner.

Fedcoin would have no impact on monetary policy

Fed officials might balk at giving the idea a shot if they feared that adopting a Fed cryptoledger would impede the smooth functioning of Fed monetary policy. They needn't worry.

The Fed currently exercises control over the price level by varying the quantity of reserves and/or the interest paid on reserves. The existence of cash doesn't get in the way of this process, nor has it ever gotten in the way. Bringing in a third liability type, Fedcoin, the quantity of which is designed to fluctuate in the same way as cash, would likewise have no impact on monetary policy. The Fed would continue to lever the return on reserves in order to get a bite on prices while allowing the market to independently choose the quantity of Fedcoin and cash it wished to hold.

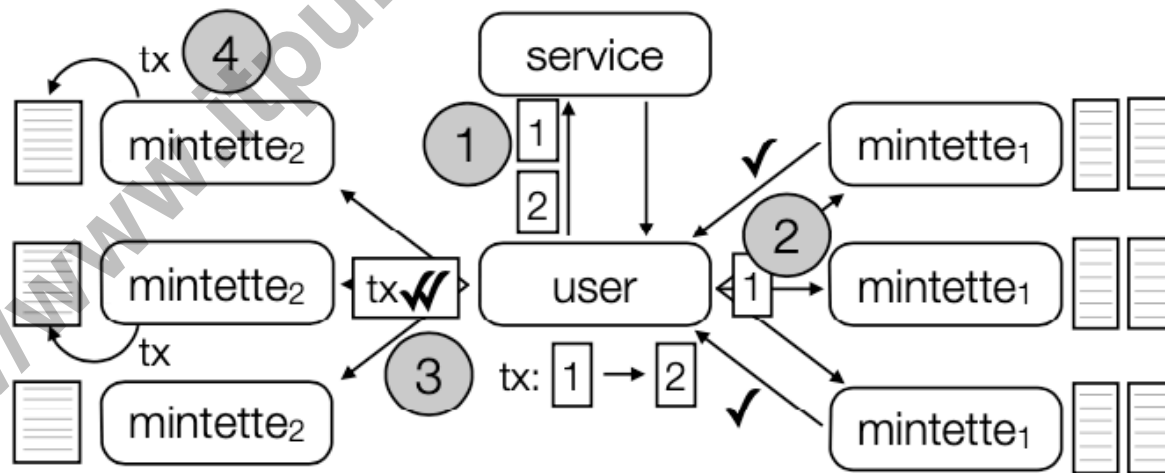
Well, almost none: Interest on Fedcoin and the zero lower bound

2014

央行学习比特币

架构融合 云化共建

2015



世界首个运行的央行数字货币

CADcoin

架构融合
云化共建

加拿大银行、加拿大支付协会、6 家商业银行、
R3 联盟

批发业务

r3.

CAD-coin versus Fedcoin

Rod Garratt

1 Introduction

June last year, details of Project Jasper were presented to the audience at Payments Panorama 2016. Though the session was closed to the press, pictures of a key slide with a reference to a digital unit of account called "CAD-coin" were tweeted to the press. The story was instantly picked up by Forbes, the Financial Times and other prominent media outlets. Why was this such big news? Was it a misunderstanding as to the true purpose of Project Jasper or was it an appropriate response to a historic event?

To answer these questions, it is useful to clarify what Project Jasper is and what many instantly perceived it to be: the Canadian version of Fedcoin.

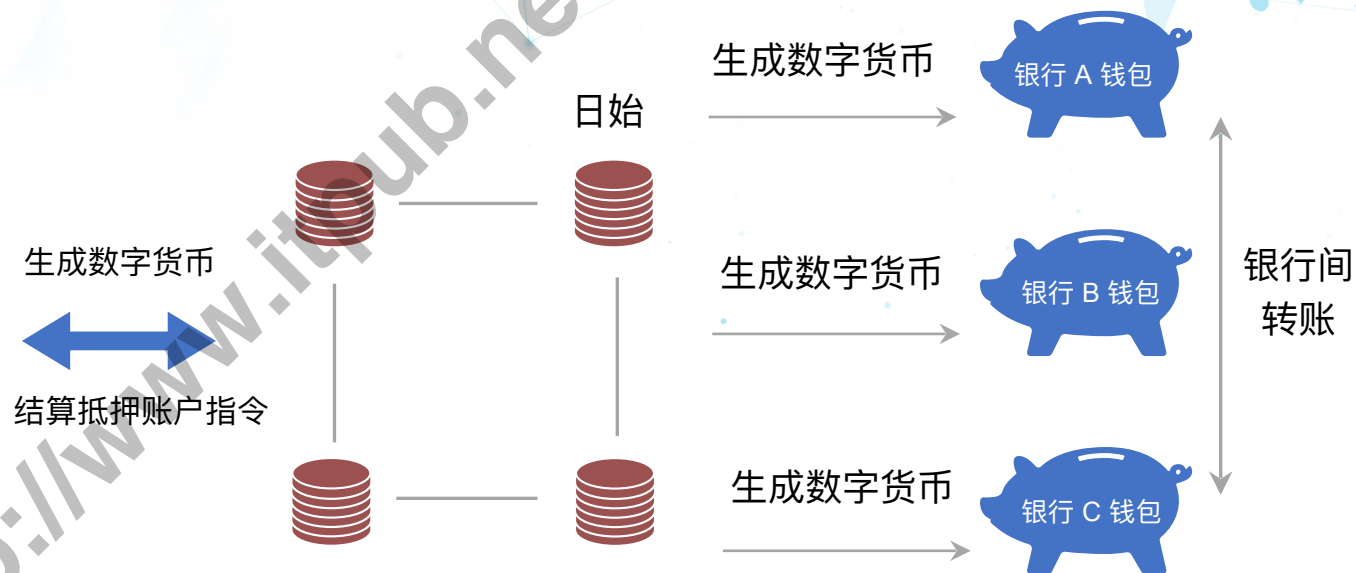
2016 清结算可以同步完成

新加坡 Ubin

新加坡电子支付系统MEPS+



去中心化 DLT



2016 交易及结算

日本-欧盟 Stella

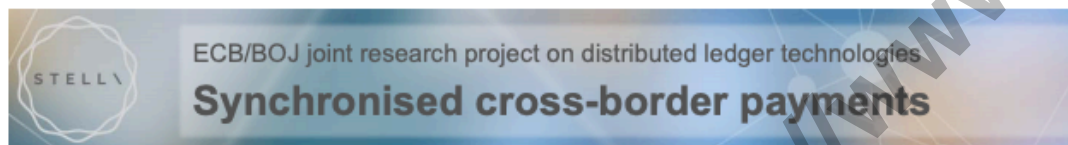
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2016

支付效率 - 证券券款对付 - 跨境支付

结算效率



Background to Project Stella

瑞典 E-krona

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E-krona

The use of banknotes and coins is declining in society. At the same time, technological advances with regard to electronic money and payment methods are proceeding rapidly. The Riksbank sees potential problems with the marginalisation of cash and has therefore initiated a pilot project to develop a proposal for a technical solution for Swedish kronor in electronic form, an e-krona. No decisions have yet been taken on issuing an e-krona.

2017

替代现金

香港-泰国 Inthanon-LionRock

架构融合
云化共建

2020年01月22日

「Inthanon-LionRock项目」的成果及下一步工作

香港金融管理局（金管局）与泰国中央银行（泰国央行）今日（2020年1月22日）公布央行数字货币(Central Bank Digital Currency, CBDC)联合研究计划—「Inthanon-LionRock项目」的成果，并发表研究报告。

作为金管局与泰国央行于2019年5月签订《谅解备忘录》的跟进工作，双方启动「Inthanon-LionRock项目」，研究CBDC于跨境支付的应用。项目下开发的泰铢—港币跨境走廊网络原型，使香港和泰国的参与银行能以点对点方式进行资金转拨和外汇交易，有助于减少多层结算程序。利用智能合约的特点，跨境资金转拨能实时完成，并实现外汇交易同步交收。

2019

外汇结算效率

厄瓜多尔 Dinero electrónico

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The World's First Central Bank Electronic Money Has Come – And Gone: Ecuador, 2014-2018



BY LARRY WHITE ON MARCH 29, 2018



2014

央行电子货币

架构融合 云化共建

央行数字货币



虚拟货币、区块链、分布式账本
2009年开始研究

央行发行通用型电子货币
1987年开始

架构融合
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3

央行 DCEP 分析

<http://www.itpub.net/>

央行论文

架构融合 云化共建

● 姚前：中国法定数字货币原型构想：2016：中国金融 - 在线与离线

- 便捷与安全

- 实名与匿名

- 交易与数据分析

- 账户与令牌

- 生态建设

- 私有云+高性能数据库+移动终端 私有云+区块链+移动终端

- 央行-商行二元体系

- 一币两库三中心

- 审慎对待智能合约

在线与离线。目前实际运转的电子货币系统基本都是在线模式。中国法定数字货币的发行与流通应能够同时支持在线与离线并行的方式。为了设计的简明，在线/离线需要有不同的规范和流程。

便捷与安全、便捷性是获得市场认可的一个重要因素，安全性则是整个体系能够健康运行的基础。便捷性与安全性之间如何权衡是一个重大问题。为了提高效率，增加便捷性，在安全机制上，大额与小额的处理有可能需要区别对待。

实名与匿名。数字货币可以实行实名制，也可以实行匿名制，也可以是两者结合。中国法定数字货币的设计考虑是“前台自愿，后台实名”。

交易与数据分析。在大数据、云计算环境下，交易安全已不完全依赖传统的身份认证体系，通过客户行为分析来保障交易安全、规避风险，这一方向值得高度关注。加强客户行为分析，是央行发行法定数字货币的重要考量。宏观上，数字货币可以做大数据分析，但微观上不可侵犯合法用户的隐私。

与银行账户的关联。目前运营的电子货币系统大都基于银行账户，而数字货币系统则不一定与银行账户关联。

生态体系建设。中国法定数字货币的设计要与金融界、科技界合作，深入研究和合理利用各种新型创新技术，以优化完善数字货币发行流通的技术框架，并充分预见技术进步，引入长期演进、不断优化的发展理念。

对区块链技术的期待。区块链技术是下一代云计算的雏形，备受各方瞩目，但作为成熟的企业级应用案例尚不多见。“私有云+高性能数据库+移动终端”与“私有云+区块链+移动终端”，有可能是两个既关联又有区别的思路。让中央更强大，让数据更安全，使终端更智能，让个人的支付行为更能动，一定是未来央行数字货币追求的目标。如果将区块链技术应用于央行数字货币的研发，是否可以对其进行必要的改造？面对大规模交易的速度和效率问题，区块链技术自身如何实现实质性突破？

第一，遵循传统货币的管理思路，发行和回笼基于现行“中央银行—商业银行”的三元体系来完成；第二，币本身的设计，运用密码学理论知识，安全性强；第三，货币的产生、流通、清点核对及消亡全过程登记，可参考区块链技术，建立集中/分布相对均衡的簿记登记中心；第四，充分运用可信计算技术和安全芯片技术来保证数字货币交易过程中的端到端的安全；第五，充分运用大数据分析技术，不仅可以进一步保障交易安全，还可以满足反洗钱等业务需求；第六，数字货币的用户身份认证采用“前台自愿、后台实名”的原则，既保证用户隐私，又规避非法交易的风险；第七，数字货币本身的设计应力求简明高效，数字货币之上的商业应用尽可能交给市场来做，同时把技术标准与应用规范做好；第八，构建由央行、商业银行、第三方机构、消费者参与的完整的数字货币生态体系，保证数字货币的发行、流通、回收全生命周期闭环可控。■

悬疑

经营范围牌照

央行采用的是“双层运营”机制。初期，央行将会把DCEP兑换给阿里、腾讯、工行、建行、农行、银联，再由他们直接向大众发行，等DCEP真正发行后，我们可以通过微信、支付宝、银行APP直接兑换。同时，央行未来还将与国外银行达成合作，从而把DCEP推向全世界。

架构融合 云化共建

接口及智能合约

[央行数字货币“DCEP对外开放接口”系假消息 | 区块链见闻](#)

2019年11月10日 - 区块链见闻,国内顶级的区块链综合信息网站、区块链投融资平台。浏览区块链新闻、解读政策法规趋势、找区块链项目投资、探知区块链最新消息、了解数...
www.qkljw.com/news/911... - 百度快照

[媒体:央行数字货币「DCEP对外开放接口」或即将公布_区块链快讯...](#)

媒体:央行数字货币「DCEP对外开放接口」或即将公布一个名为「DCEPAPI」
(http://Vdcepapi.com/)的网站悄然出现在大众视野。网站内容写有:「努力...
www.edenw.com/live/30... - 百度快照

[看来是真的!突发:央行数字货币「DCEP对外开放接口」或即将重磅...](#)

2019年12月10日 - 突发:央行数字货币「DCEP对外开放接口」或即将重磅公布 YttHxnOO发现一个名为「DCEPAPI」(http://dcepapi.com/)的网站悄然出现在大众视野。网站内容写...
[全球五金网](#) - 百度快照

[突发:央行数字货币「DCEP对外开放接口」即将重磅公布 - 格时财经](#)

2019年11月9日 - 「DCEP支付接口即将开放」、「请持续关注我们,将有重磅消息公布」、「...日消息,记者获悉,央行旗下金融科技研究院正批量招聘人员,参与法定数字货币...
www.gscaijing.com/?p=3... - 百度快照

[央行数字货币“DCEP对外开放接口”系假消息_区块链_建材之家JC68...](#)

2019年11月10日 - 1.央行数字货币“DCEP对外开放接口”系假消息 该网站未备案 2.雄安新区区块链创新实验室今日宣布成立 3.华为张小军:华为区块链未来发展方向是打造区块链...
<https://www.jc68.com/qkl/show-...> - 百度快照

可以基于 DCEP 做什么？

功能及目标

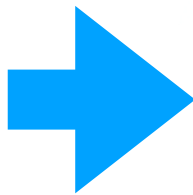


DC 兑换

DC 查询

钱包管理

钱包挂靠



普惠金融

用户隐私

离线支付

无假币

克服
纸钞缺陷

反洗钱

可追溯

可编程

货币主权

人民币
国际化

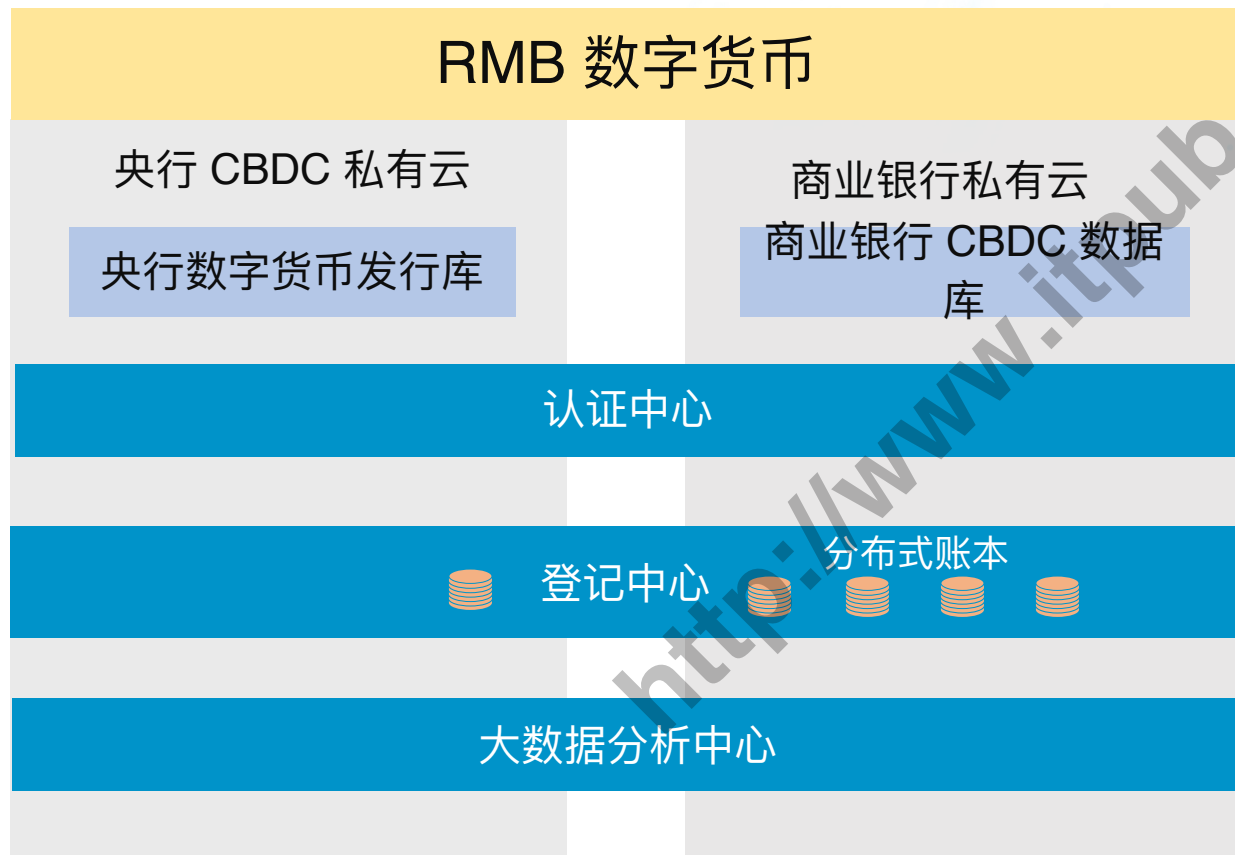
架构融合
云化共建

架构融合
云化共建

4

央行 DCEP 架构

一币两库三中心



登记中心:

- 发行登记 CBDC
- CBDC 用户身份
- 权属登记
- 记录交易流水
- 核销 CBDC
- 网站查询
- 脱敏后写入分布式账本

原理

中心化系统

可运行在央行，也可在商业银行，但必须符合央行规范

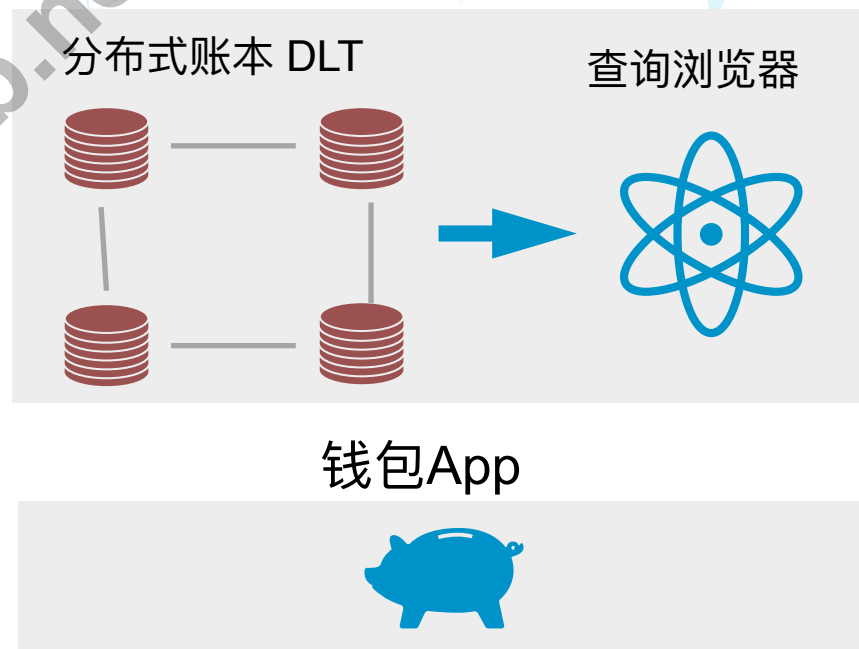


架构融合

云化共建

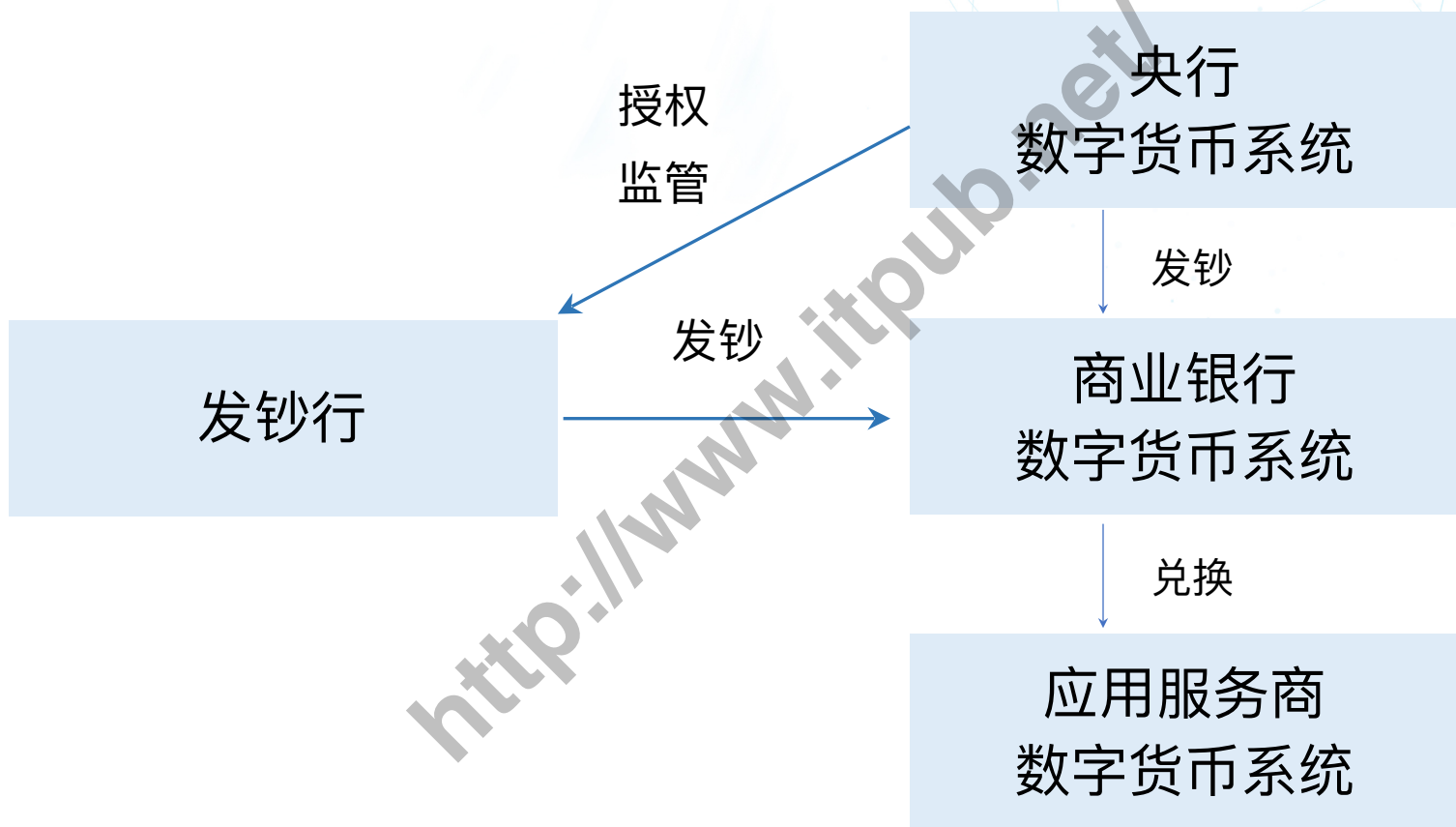
去中心化 DLT

验钞机



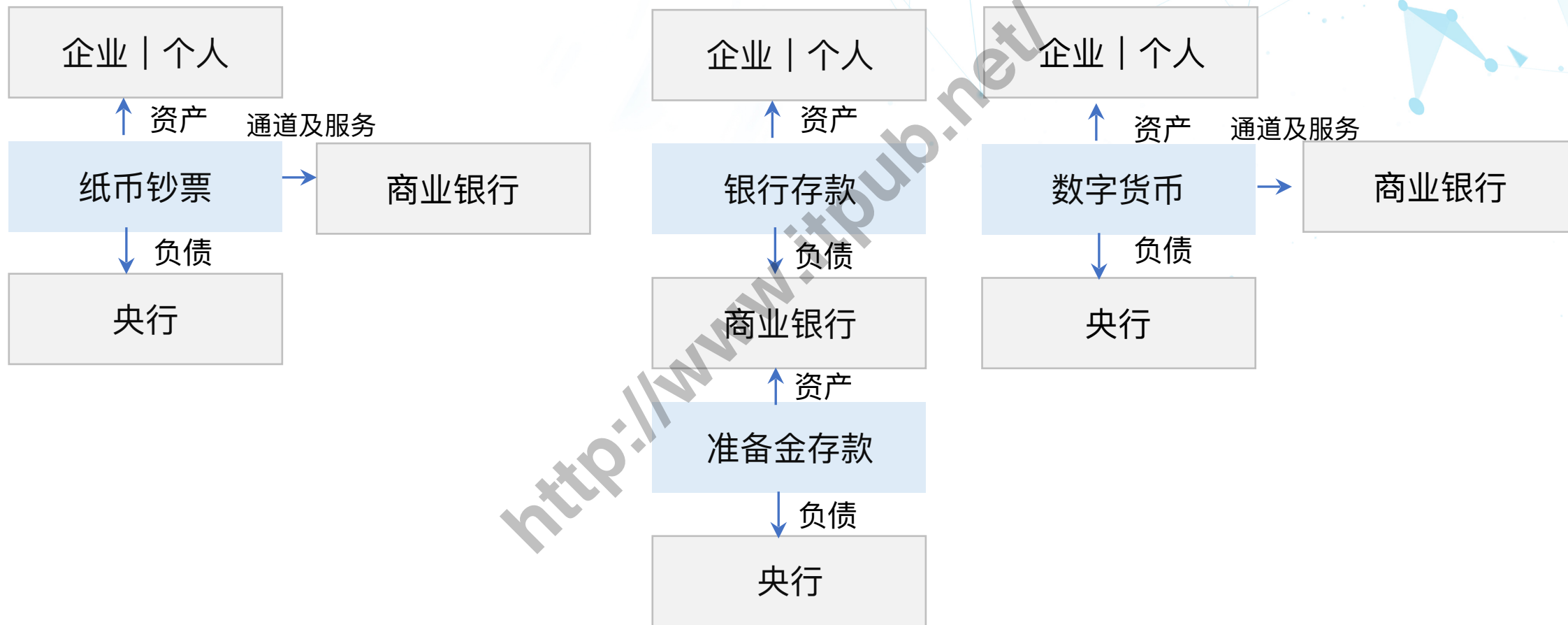
二元体系-三层结构

架构融合
云化共建



数字货币经济关系

架构融合
云化共建



数字货币字段

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云化共建

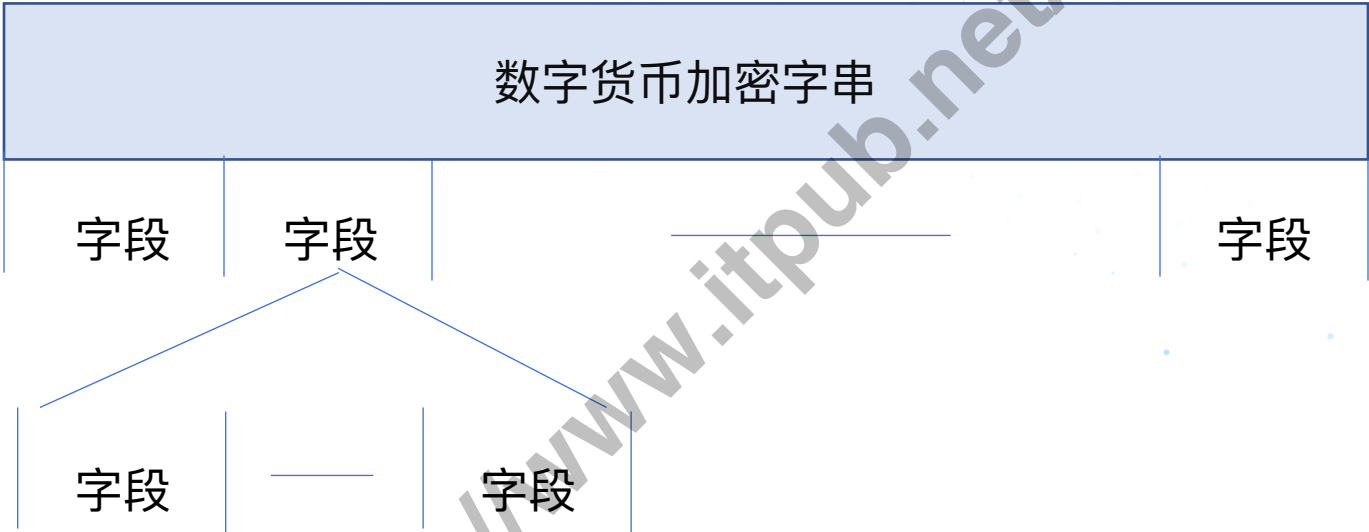
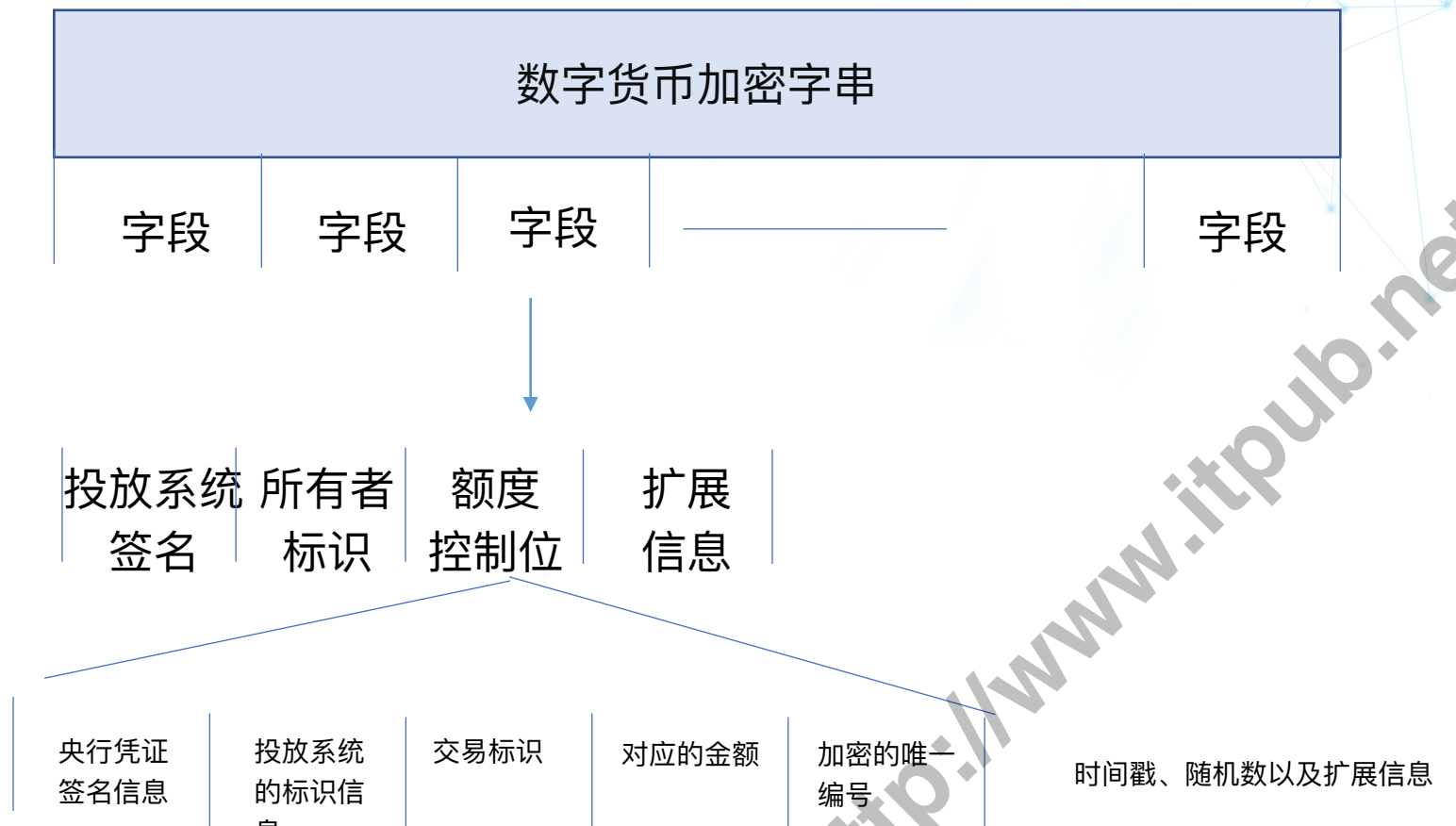


图 1

发行方 标识	所有者 标识	货币管 理属性	货币安 全属性	货币应 用属性	数字货币 加密字符串
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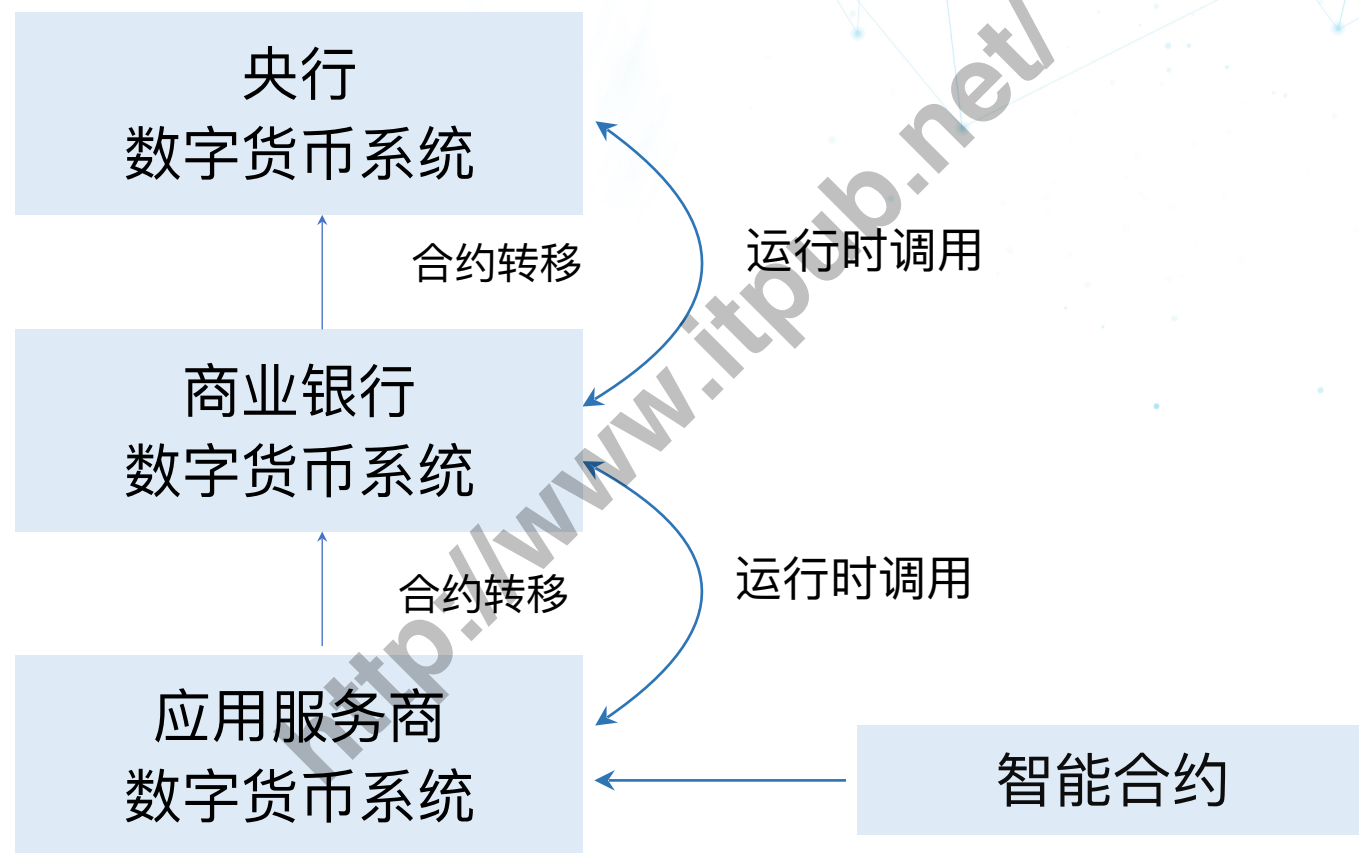
架构融合 云化共建

2020

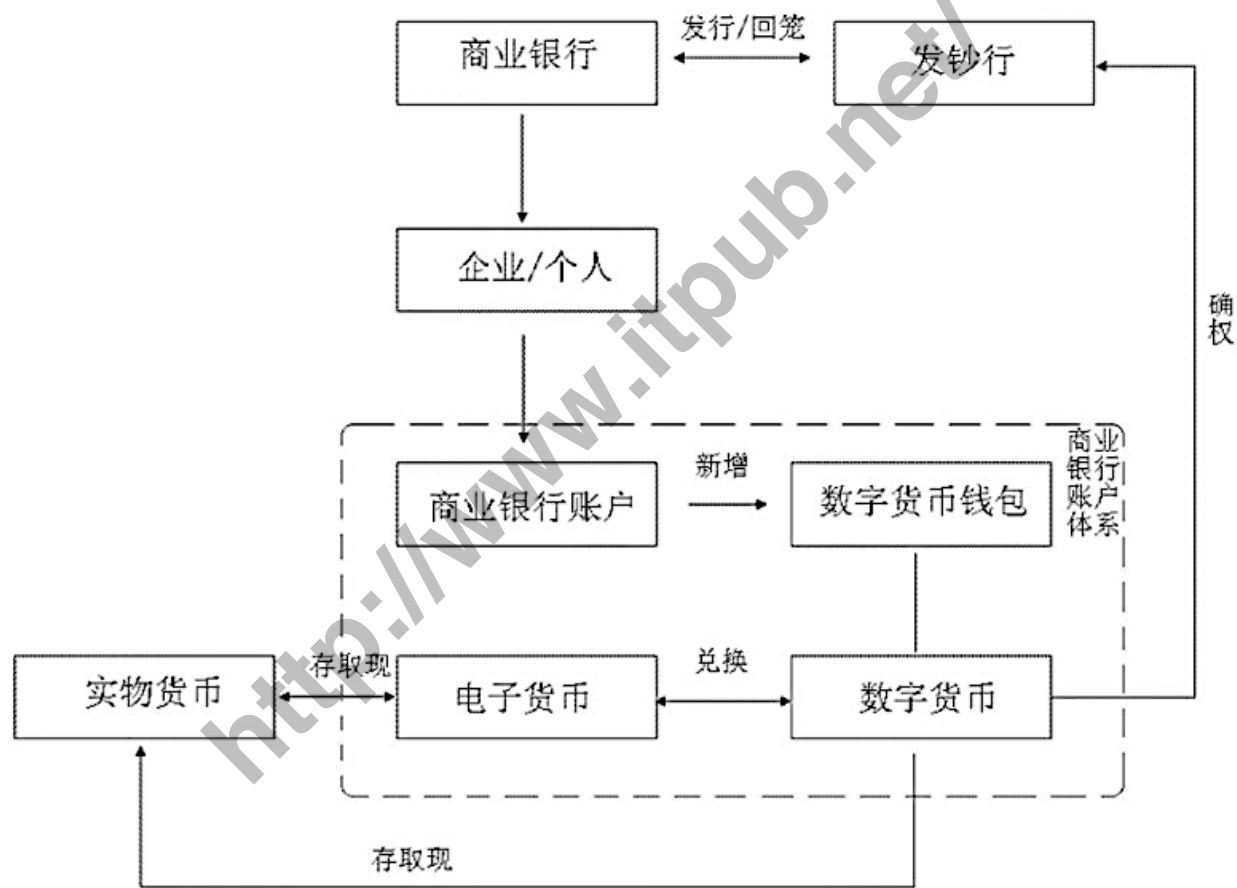


智能合约

架构融合
云化共建



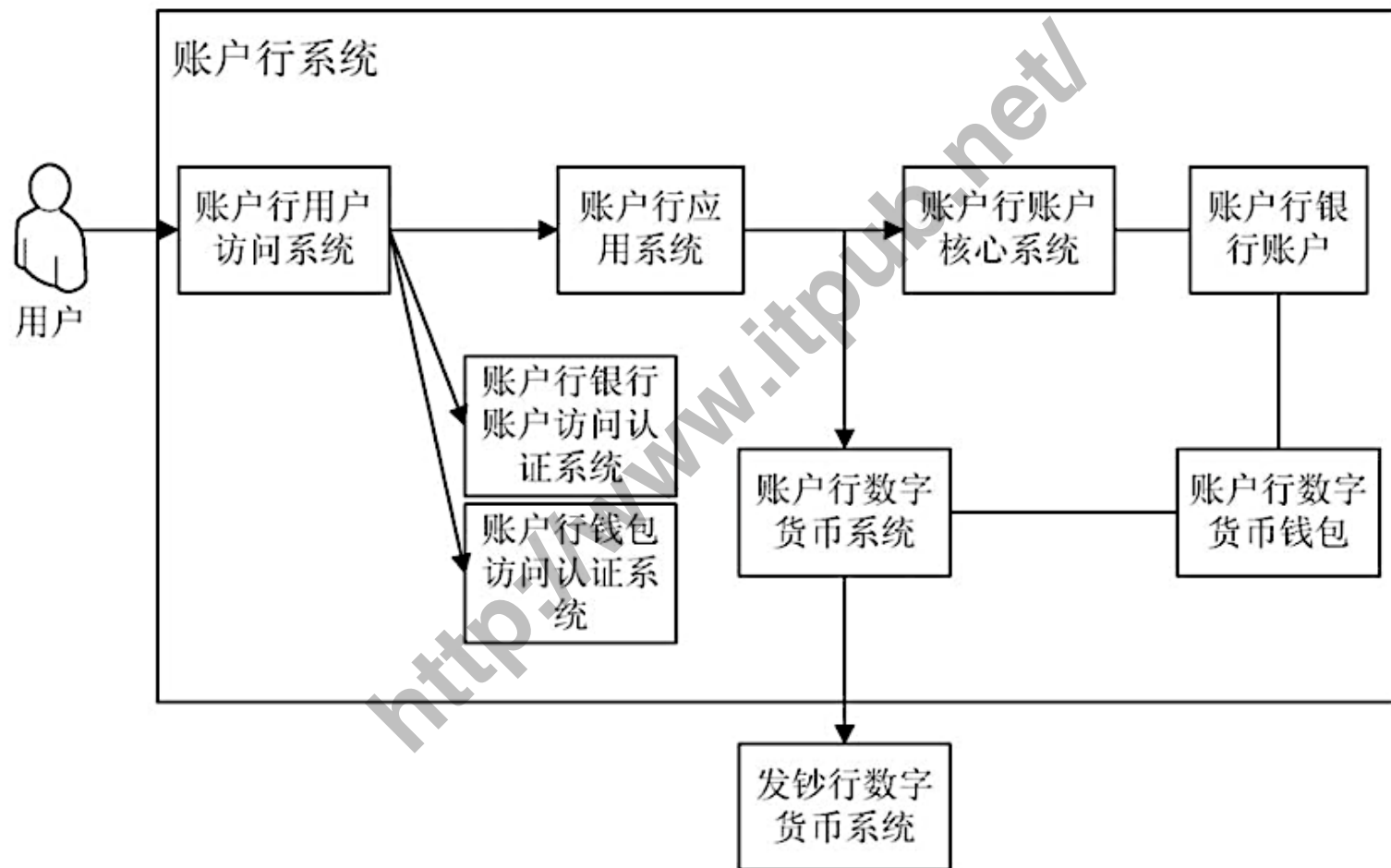
数字货币管理体系架构图



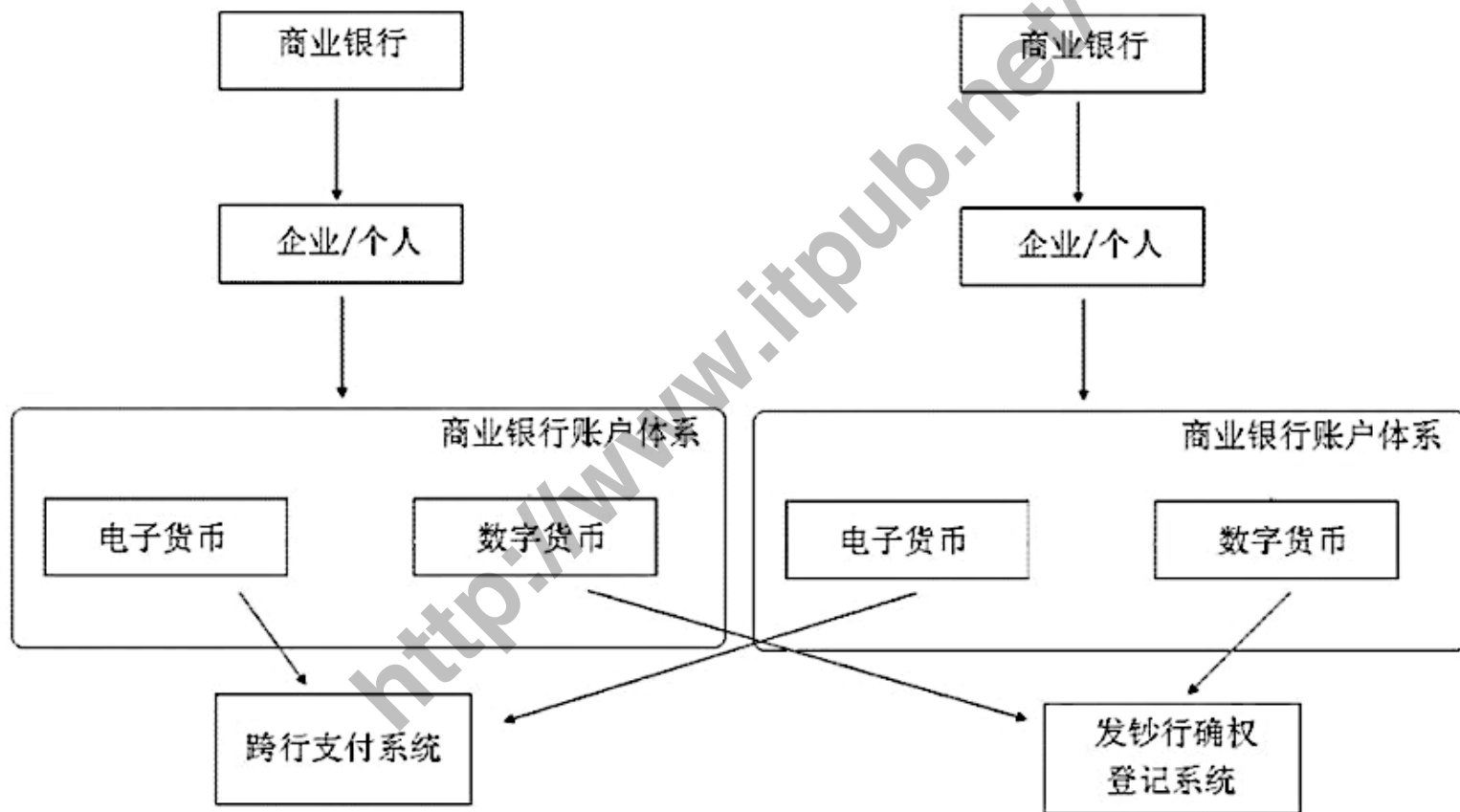
架构融合

云化共建

数字货币管理总体架构图

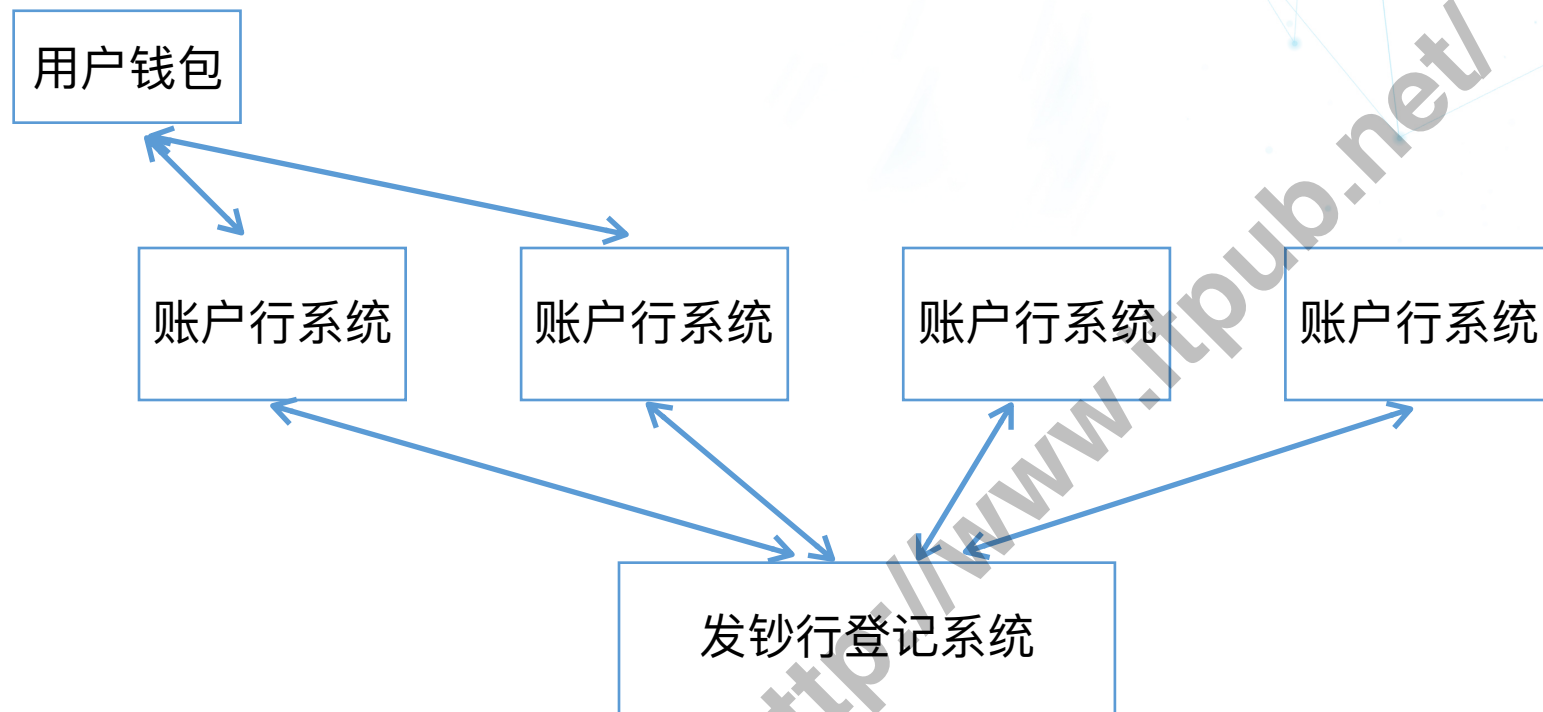


数字货币交易通道架构图



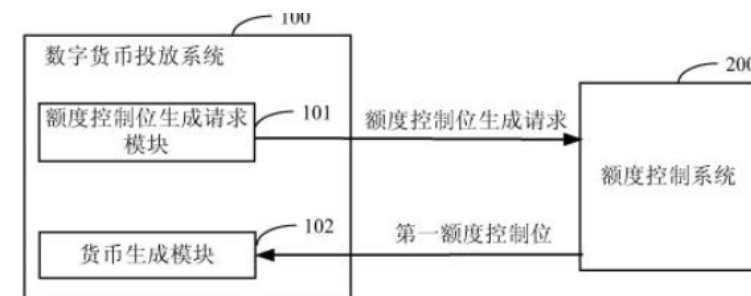
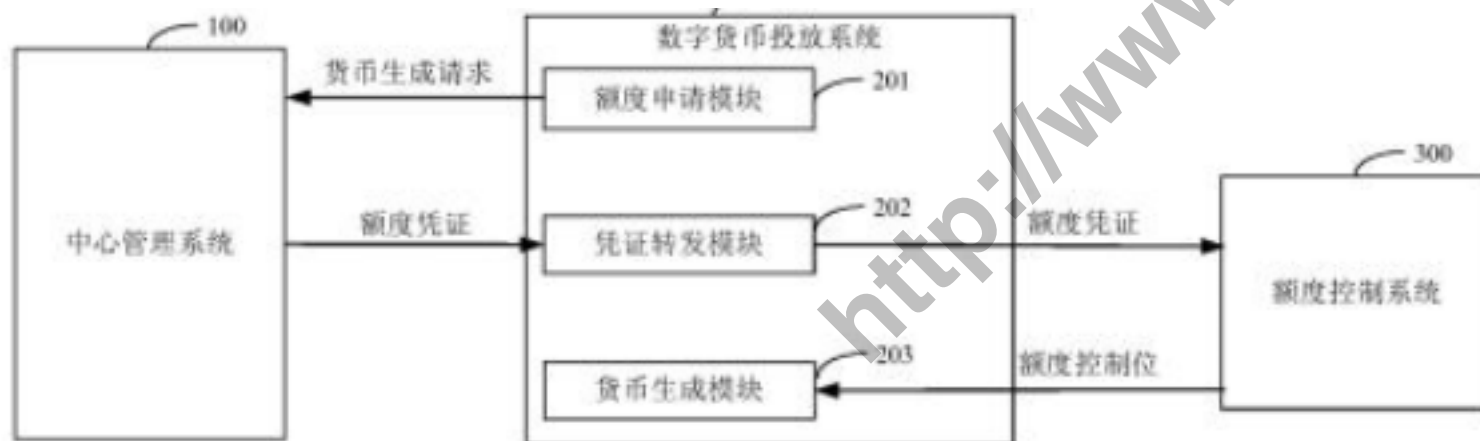
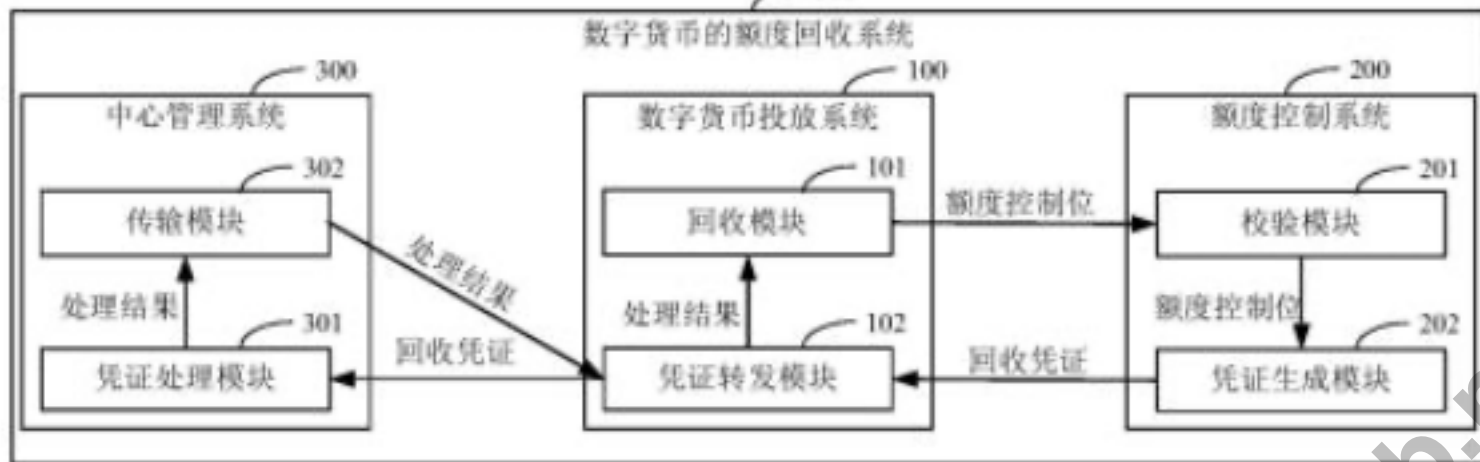
分布式

架构融合
云化共建



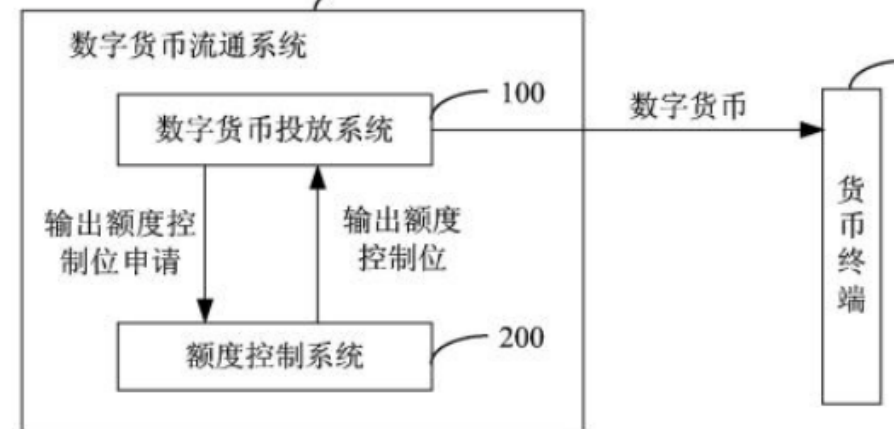
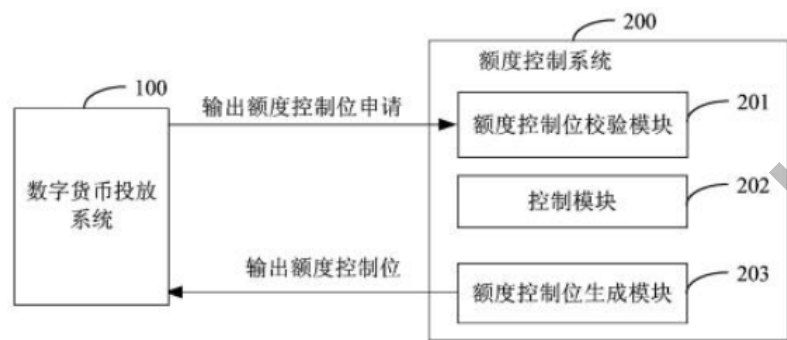
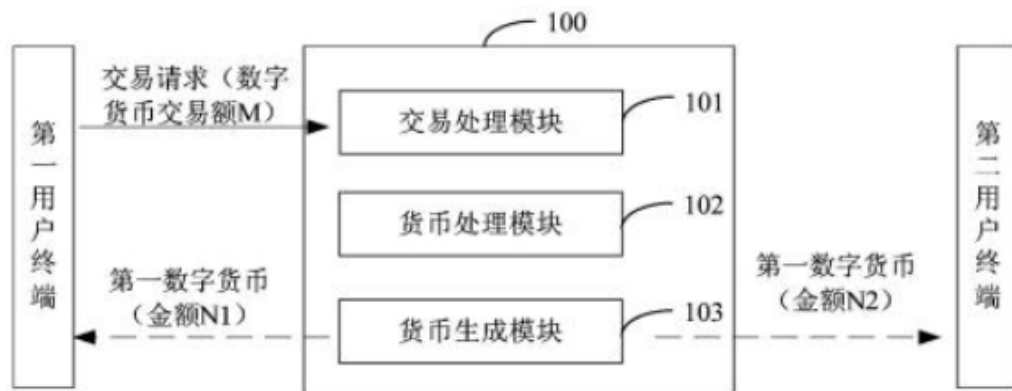
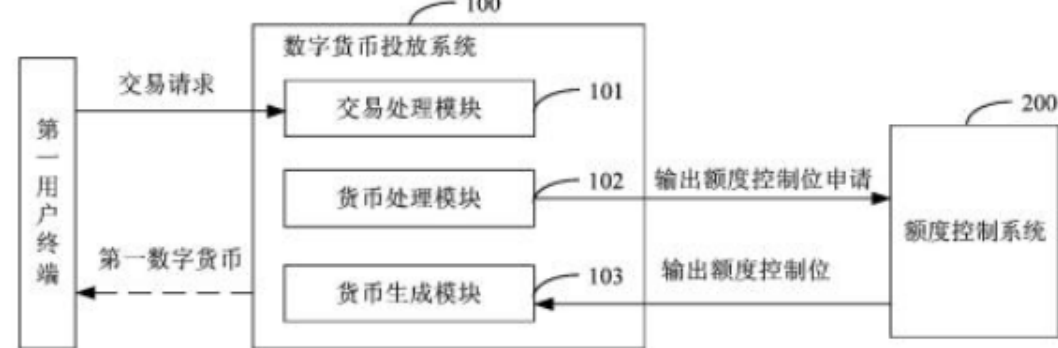
架构融合 云化共建

2020

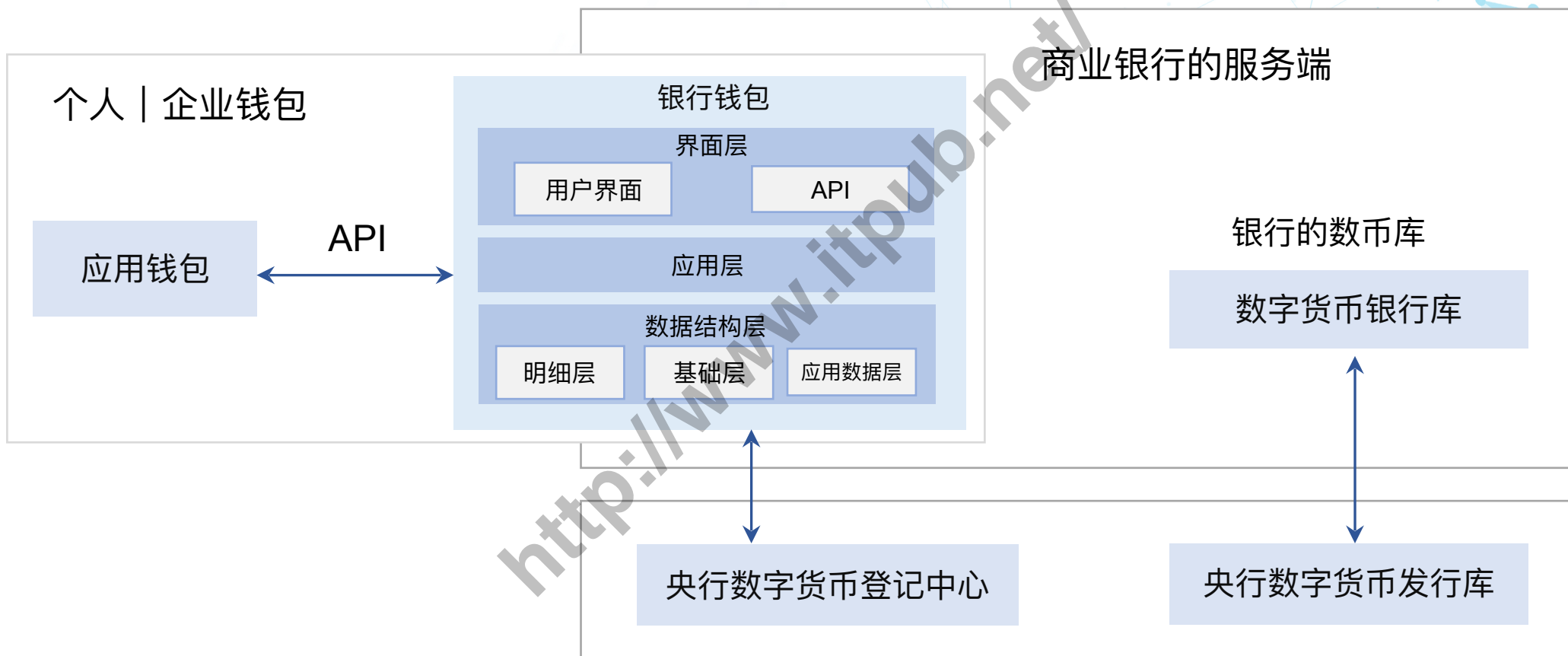


架构融合 云化共建

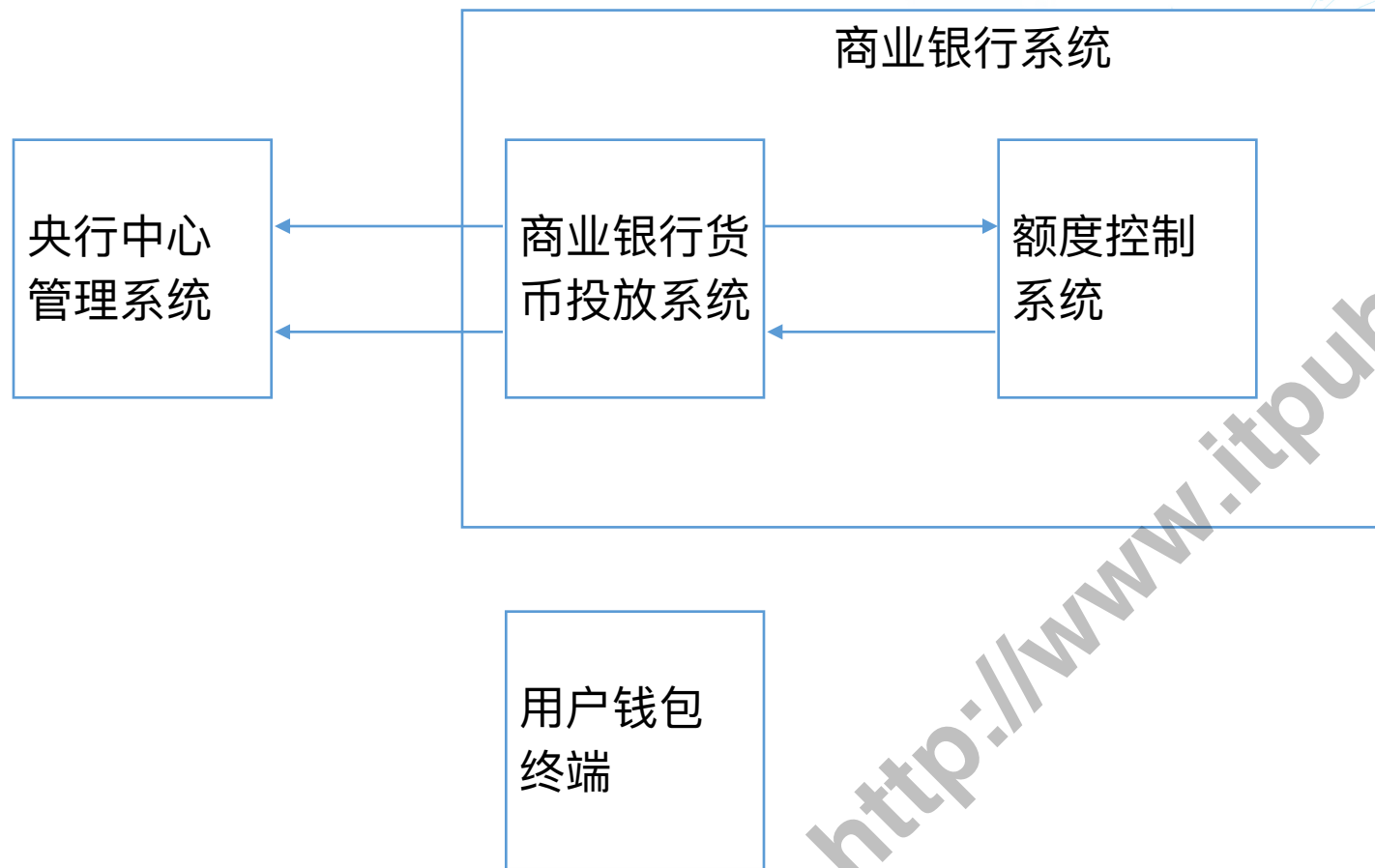
2020



钱包与库结构



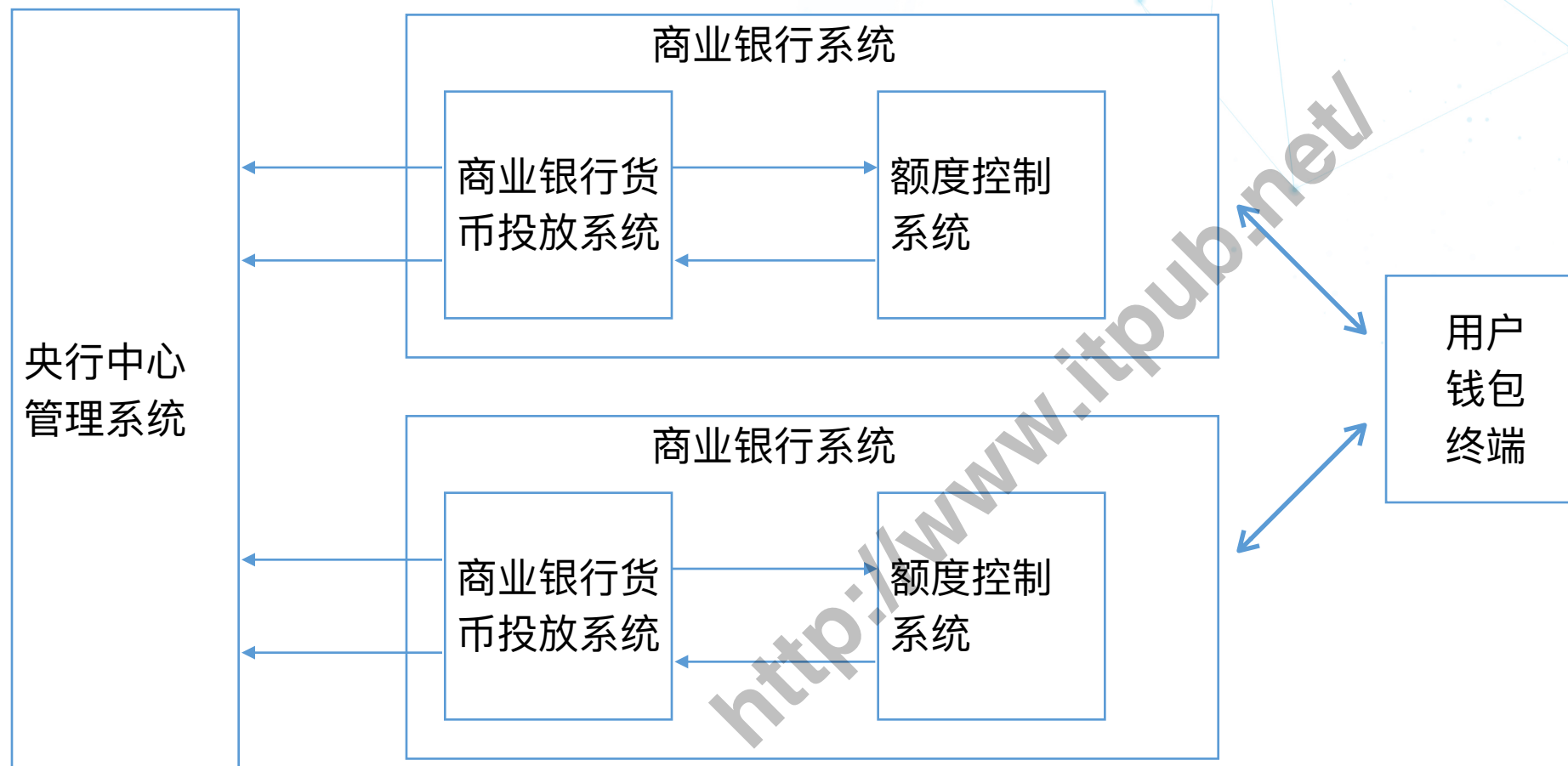
架构融合 云化共建



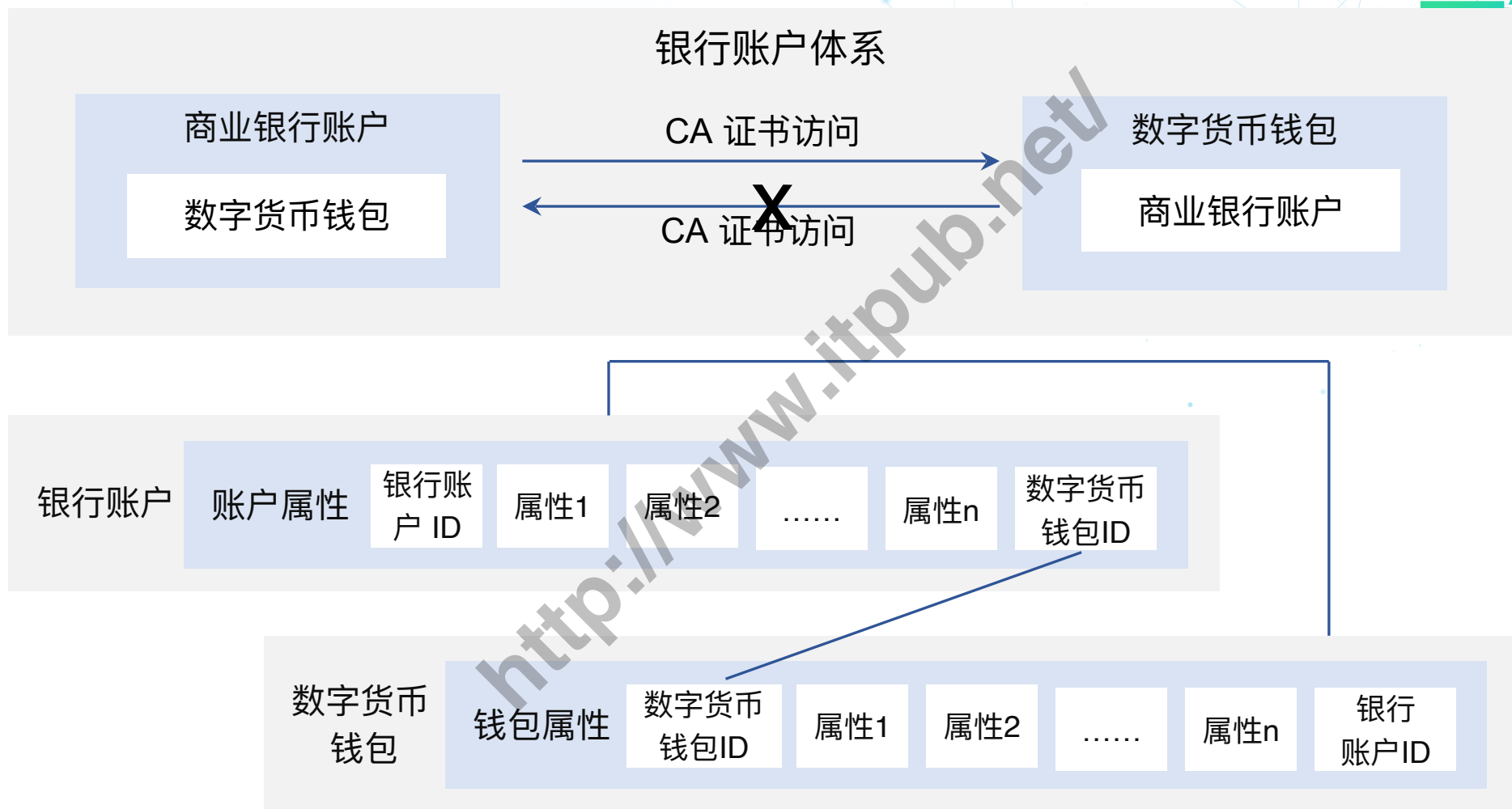
2020

额度管理

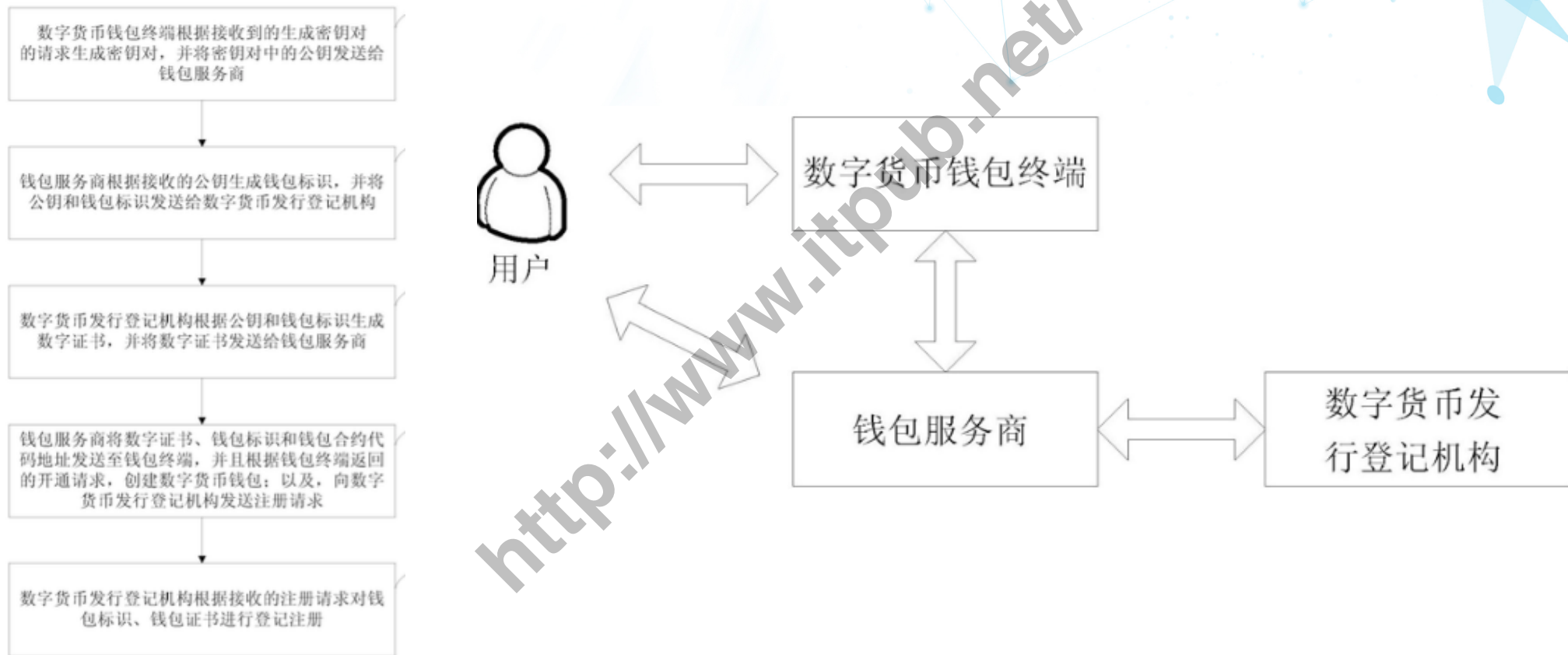
架构融合 云化共建



架构融合 二化共建



开通数字货币钱包



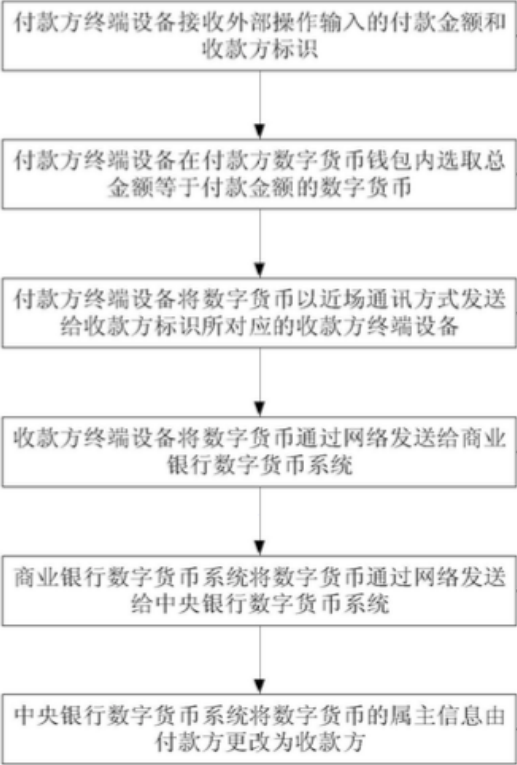
支付方式

- 不是账户余额
- 也不是 UTXO
- 是登记作废原有的来源币字符串，生成去向币字符串



如通过记录输入数字货币转换为输出数字货币的形式完成对输入数字货币的注销，或将输入数字货币转换为输出数字货币的记录上报给管理数字货币投放系统的中心管理系统

后台登记方法



数字货币名	属主	备注
Pbc100adfk109987766670	138xxxxx 001	D币100
.....		
Pbc50cadfk109987766670	137xxxxx 002	D币50

数字货币名	属主	备注
Pbc100adfk109987766670	1Xadcfdgdadg	D 币 100 H (138xxxxx001)
.....		
Pbc50cadfk109987766670	2xcfdald3xgdf	D 币 50 H (138xxxxx002)

几种转移方法

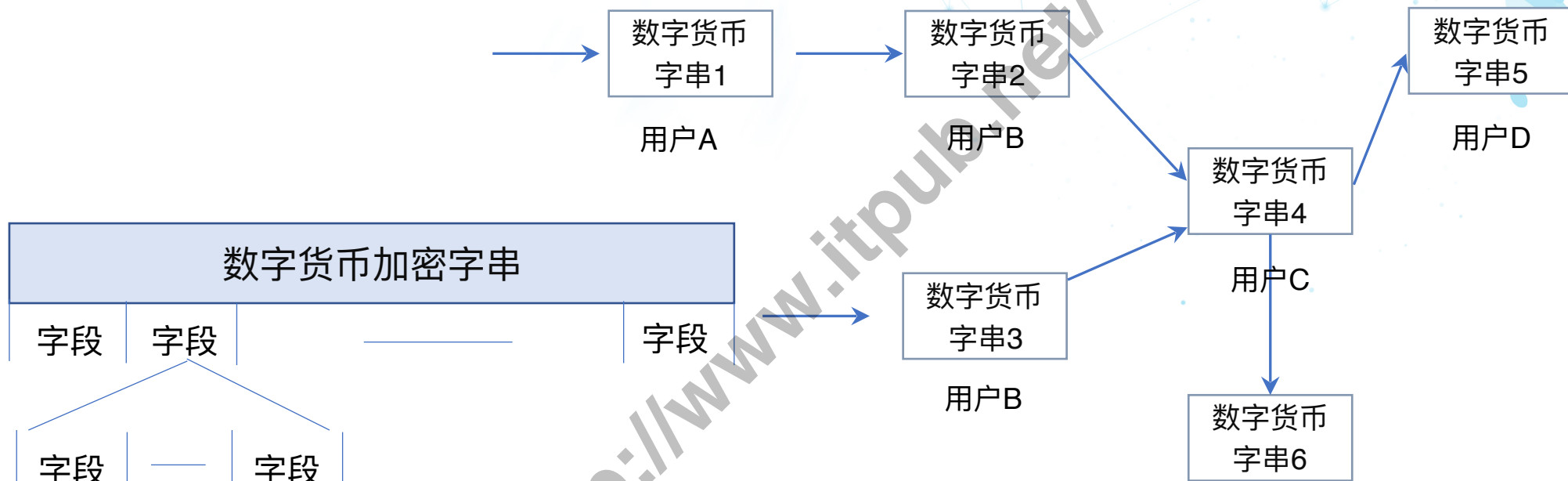
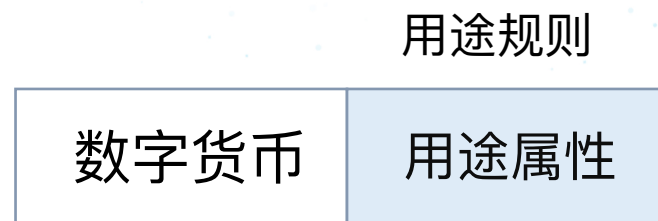
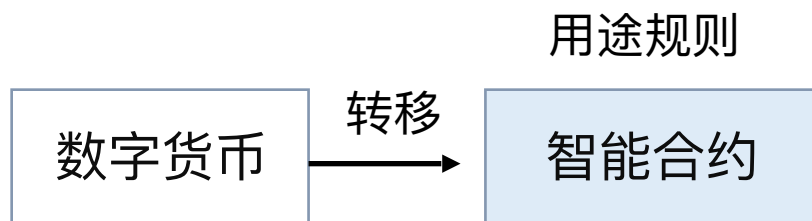


图 1

直接转移 合并转移 拆分转移

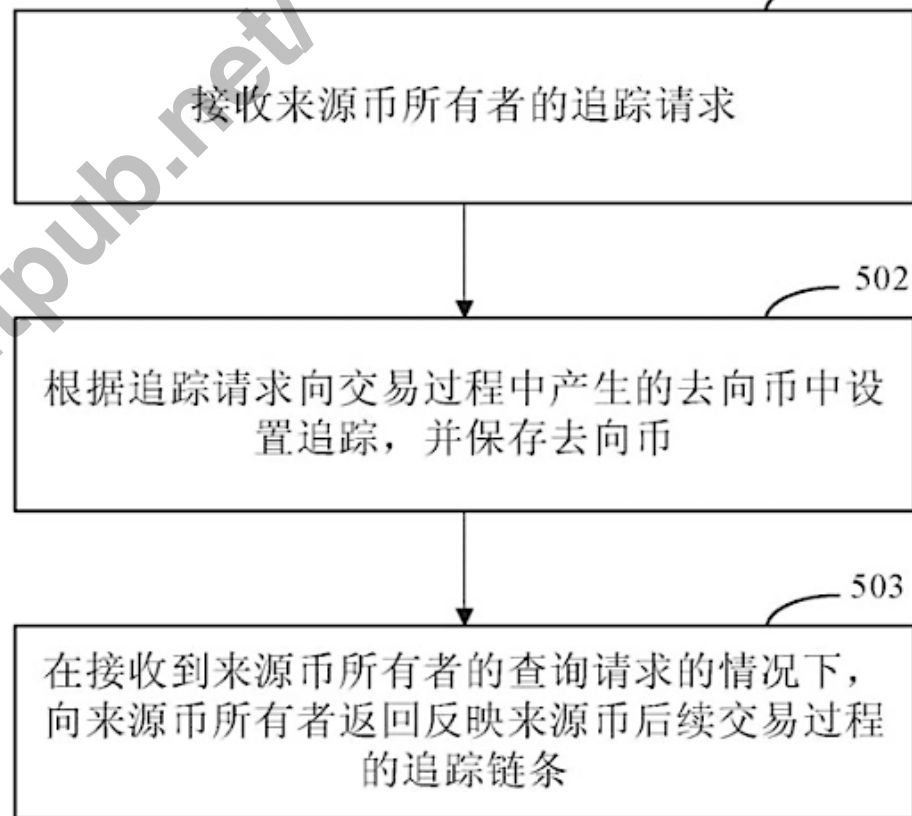
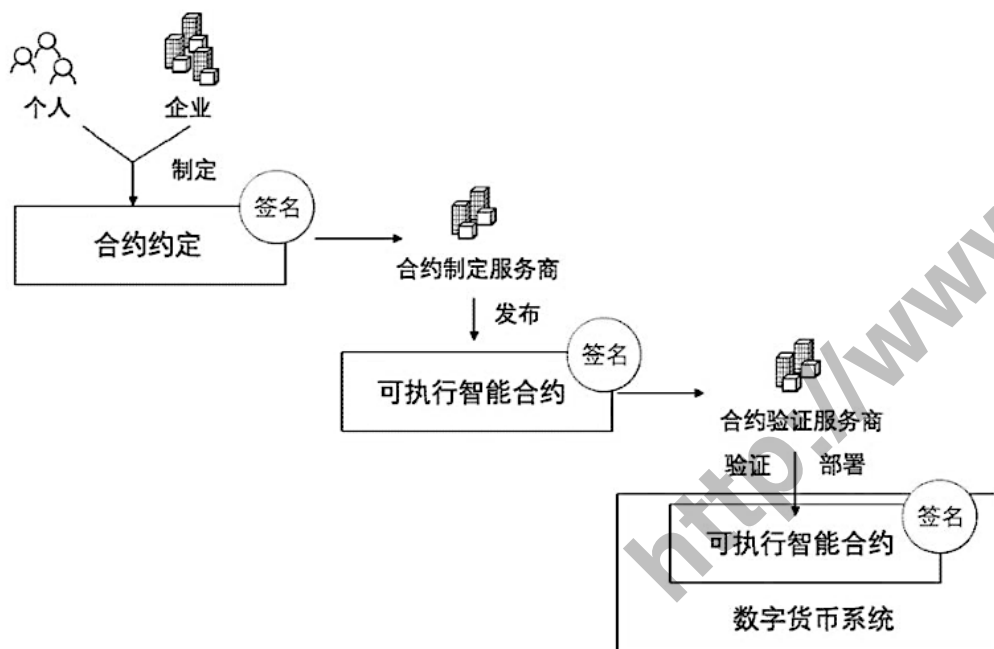
定向使用数字货币的两种方式

- 在智能合约中约定用途规则
- 在数字货币的用途属性中约定



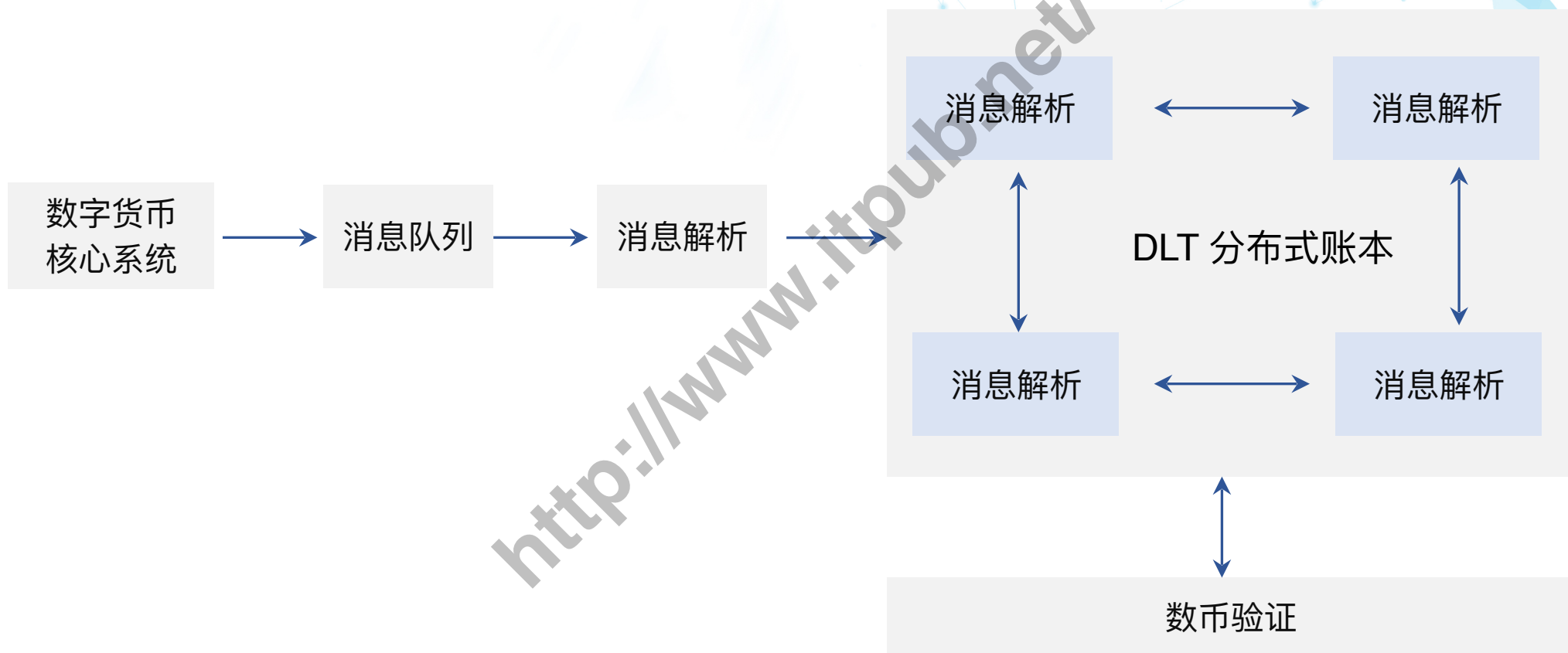
- 智能合约中登记数字货币串
- 数字货币串指定智能合约

可追溯的实现



可追溯的实现

架构融合
云化共建



架构融合
云化共建

Thanks

<http://www.itpub.net/>